

Kitchen STEAM

Forty-six activity cards

One card per page. Every card carries two things the first deck did not: **Everyone participates**, which says how a child who is not going to taste anything does the whole session; and **Across K-5**, which says who does what in a mixed-age room. The explanation marked *for the adult* is for you — hold it until the children have predicted.

SENSES & SELF 5 CARDS

N01	The Taster's Alphabet	Building a sensory vocabulary the whole class shares	classroom	45 min, then five minutes every week after
N02	Eight Ways to Meet a Food	The class agreement, taught to everybody on day one	classroom	40 min, in the first session
N03	Listen to It First	Sound as a food property, and the easiest door in	classroom	40 min
N04	Three Cups, One Difference	The triangle test — how professionals ask whether a difference is real	kitchen	50 min
N05	A Table Where Everyone Fits	Hosting, asking for what you need, and minding your own plate	kitchen	55 min

SCIENCE 8 CARDS

S01	The Cucumber Weeps	Osmosis — water moving where you did not put it	kitchen	50 min plus a 30-minute wait
S02	Pulling Colors Apart	Capillary action and paper chromatography	kitchen	55 min
S03	Three Ways to Set a Liquid	Gelatin, agar and pectin — the gels do not behave alike	kitchen	60 min plus setting time
S04	The Apple Goes Brown	Enzymatic browning, and four ways to stop it	kitchen	45 min plus checking back
S05	Your Tongue Is Not My Tongue	How plants defend themselves, and why people taste them differently	kitchen	55 min
S06	Four Families of Color	What heat and acid do to the pigments in plants	kitchen	55 min
S07	Closer Than You Have Ever Looked	Magnification, crystals and the structure of ordinary things	classroom	50 min
S08	The Cracker Turns Sweet	Digestion as a system — mechanical and chemical	classroom	45 min

ART 10 CARDS

A01	The Color in the Peel	Natural dyes, mordants and colorfastness	kitchen	90 min over two sessions
A02	A Color Wheel You Can Eat	Primary, secondary and complementary color; in produce	classroom	50 min
A03	The Okra Is a Stamp	Relief printmaking, repeat pattern and the edition	classroom	70 min
A04	Draw It Before You Know It	Observational and botanical illustration	classroom	60 min
A05	The Plate Is a Frame	Composition — thirds, odd numbers, negative space, line	kitchen	55 min
A06	Where Is the Light Coming From?	Photographing food — direction, diffusion, angle	classroom	55 min
A07	Dough Is a Material	Sculpture — form, volume and the thing that holds its shape	kitchen	80 min plus baking
A08	Round and Round and Round	Radial symmetry, repeat units and pattern	classroom	55 min
A09	The Kitchen Orchestra	Sound, rhythm and a score anyone can read	kitchen	50 min
A10	What Shall We Call It?	Naming, typography and the identity of a dish	classroom	60 min

ENGINEERING 6 CARDS

E01	A Tool for This Hand	Designing for a real person, not an average one	kitchen	80 min
E02	Which Shape Holds the Sauce?	Pasta geometry, surface texture and a measurable test	kitchen	70 min
E03	The Lunchbox Problem	Structure, moisture and a sandwich that survives until noon	kitchen	60 min plus a four-hour wait
E04	What Kneading Actually Does	Dough as a material, and a test you can hold up to the light	kitchen	75 min
E05	Making Water Clear Is Not Making It Safe	Filtration, and the difference between clean and clear	kitchen	70 min
E06	Design This Kitchen	Workflow, stations and how far a person has to walk	kitchen	65 min

TECHNOLOGY 5 CARDS

T01	Twelve Pictures a Second	Stop-motion animation of a recipe	kitchen	90 min over two sessions
T02	Ask, Then Stop Talking	Recording an interview about a food that matters to someone	classroom	60 min plus interviews at home
T03	The Cold Chain	Thermometers, data logging, and the gap food must not sit in	kitchen	55 min plus a logging day
T04	Make the Book	Layout, editing and publishing the class cookbook	classroom	Three sessions across the term
T05	Before Anyone Owned a Thermometer	How people cooked precisely without precise instruments	kitchen	55 min

MATH 6 CARDS

M01	Count the Spirals	Fibonacci numbers, growing things, and a pattern nobody planned	classroom	55 min
M02	Cooking for Two Hundred	Scaling up, division with remainders, and what a tray holds	classroom	60 min
M03	The Potato in the Jug	Volume of a lumpy object, by displacement	kitchen	50 min
M04	Watching Something Cool Down	Line graphs, rate of change, and a curve that flattens	kitchen	55 min
M05	The Class Vote	Anonymous data, fair comparison, and what a number can honestly claim	kitchen	45 min
M06	A Pinch, a Handful, a Gram	Non-standard and standard units, and why anyone bothered	classroom	50 min

TABLE & COMMUNITY 6 CARDS

C01	The Dish That Traveled	A recipe with a person attached to it	kitchen	Two sessions plus home time
C02	One Grain, Many Kitchens	The same staple, cooked completely differently	kitchen	70 min
C03	The Part You Were Going to Throw Away	Whole-ingredient cooking, stock, and what waste actually is	kitchen	65 min
C04	Everyone Who Touched It	The people between a field and a plate	classroom	60 min, plus a visit if you can arrange one
C05	Cook It for Someone Else	A real recipient, a real delivery	kitchen	Two sessions
C06	How the World Eats	Utensils, table shapes and the rules nobody wrote down	kitchen	55 min

The Taster's Alphabet

Building a sensory vocabulary the whole class shares

How many words can you find for what this feels like, before anyone says whether they like it?

Grades K-2 & 3-5 45 min, then five minutes every week after Light mess No heat needed Classroom is fine

YOU WILL NEED

- A long strip of paper across one wall, divided into five columns: Looks · Sounds · Smells · Feels · Tastes
- Cards and thick markers
- Four or five contrasting items — a cracker, a grape, a slice of bread, a piece of hard cheese, a lemon wedge
- A starter word list for the adult, not posted
- A tray, so that items can be examined without being handed round

WHAT YOU DO

1. Introduce the one rule: this week we are collecting words, not opinions. 'Good' and 'yuck' are not words we are collecting.
2. Take one item. Everyone gives one word for how it LOOKS. Write every word on the wall. No word is rejected.
3. Do the same for sounds — shake it, tap it, break it near your ear. Then smell.
4. Feel: with fingers, or with a fork, or by looking at someone else doing it. All three count.
5. Taste words go up at the same time as the others — from the adult's starter list, from anyone who wants to add one, and on written slips as well as out loud. Nobody is asked around the circle for a taste word.
6. From then on, five minutes at the start of every session: add words, and star the ones nobody had last week.

EVERYONE PARTICIPATES

Every child contributes words from wherever they are. The looking and listening columns fill up first and they are full participation, not a lesser version of it. Nobody is asked whether they tasted, ever, and the wall does not record who said what — it belongs to the class. Collect all five columns on written slips rather than around the circle, so no column has a different participation shape from any other — a child who passes on one round in a round-robin is visible to everybody, and in week one that is the whole ballgame.

WORTH ASKING

- Is there a better word than that one? What is the difference between the two?
- Which column is hardest to fill? Why do you think that is?
- Did anybody find a word nobody else had?

JOURNAL The word bank pages — these fill up across the whole course

WHAT IS ACTUALLY HAPPENING · FOR THE ADULT

This card is the spine of the whole course, and it is worth understanding why. Sensory food education — the approach developed as the Sapere method and now embedded in several national early-years frameworks — moves children deliberately from *I like it* / *I don't like it* to description. Studies of sensory education with 8–12 year olds have found reduced food neophobia and increased willingness to try unfamiliar foods, and classroom studies have found that groups doing sensory work do not show the peer-avoidance drift that ordinary groups do. Refusal, it turns out, is contagious; description is not.

For this class the structural point matters more than the evidence. **If the assignment is a description, a child who only looked and smelled has completed it.** Nothing has to be adapted, nobody is given a different task, and there is no moment at which some children succeed and others do not. That is universal design rather than accommodation, and it is why this card comes first.

Keep the wall clear of negative-disgust words — rotten, revolting, gross. They are real vocabulary and in a group they license disgust-signalling, which is the one thing that makes a room unsafe fast.

ACROSS K-5

Younger children supply the vivid words and older children supply the precise ones, and both are needed. Give grades 4–5 the scribing job and require them to write down a kindergartener's word exactly as said. A five-year-old's 'it sounds like walking on snow' is better than 'crunchy' and the older children should be told why.

EXTEND IT Play the description game: one child describes something hidden using sensory words only and the others identify it. Grades 4–5: build an intensity scale for one property — rank five foods from least to most crunchy and defend the order.

WATCH OUT The rule has to be held from minute one. The first time a child says 'ew' is the moment that decides what kind of room this is — redirect immediately and without heat: 'give me a word for what it's actually like.'

THE THREAD UNDERNEATH Describing something takes longer than judging it, and you end up knowing more.

Eight Ways to Meet a Food

The class agreement, taught to everybody on day one

What counts as meeting something new?

Grades K-2 & 3-5 40 min, in the first session Light mess No heat needed Classroom is fine

YOU WILL NEED

- A large printed wheel with eight ways around it: Be near it · Look · Listen · Smell · Move it · Touch it · Lip it · Taste it
- A small plate for every single place setting, labeled 'not for me today'
- One unfamiliar but unthreatening item per child — a fresh herb sprig works well
- The class agreement, printed large, to be signed or handprinted by everyone

WHAT YOU DO

1. Show the wheel. Say clearly: these are eight ways to meet a food, and they all count. There is no top and no finish line.
2. Everyone does 'be near it' together. That is one. Everyone has now met the food.
3. Go round the wheel as a group, doing each one that people want to do, in any order the class picks.
4. Introduce the 'not for me today' plate at every setting, including the adults'. Practice using it out loud.
5. Agree the rule that makes it work: nobody comments on what anyone else eats. Practice that too, with a role-play.
6. Everyone who wants to signs or handprints the agreement — signing is optional, and the adults sign first, because it is the room's commitment before it is anybody's. Post it where it can be pointed at without a word being said.

EVERYONE PARTICIPATES

This card is the mechanism by which every other card in the deck includes everybody, so run it in session one and refer back to it constantly. Every plate gets the small plate; every adult uses the phrase at least once in front of the children; the wheel stays on the wall all term.

WORTH ASKING

- Which way did you meet it today?
- What should we say when someone says 'not for me today'?
- Why do you think we agreed nobody comments on anybody else's plate?

JOURNAL The ways-to-meet page — circle any that you did, in any order

WHAT IS ACTUALLY HAPPENING · FOR THE ADULT

Feeding therapists work with a long graduated hierarchy — tolerating a food in the room, then at the table, then interacting with it, smelling it, touching it with a progression that works inward from fingertip to hand to face to lips, then tasting, then eating. It runs to more than thirty steps, and roughly the first half involves no contact with the mouth at all. One of those middle bands is, almost word for word, *helping prepare it, stirring it, serving it to other people*. **A cooking class is already most of that ladder.** You do not need to add a tasting requirement to be doing real exposure work; you need to run a kitchen and get out of the way.

Two deliberate departures from the clinical version. It is drawn as a **wheel**, not a **ladder**, and it is not numbered — a ladder implies a top, a top implies falling short, and in a group that converts a private step into a public ranking. And no record is kept of where any child is. Building an individual child's hierarchy is clinical work that belongs to their own team; what a classroom can offer is an environment where the bottom two-thirds of that ladder happen constantly and unremarkably.

The 'not for me today' wording is doing real work. It is time-limited, it needs no reason, and it leaves tomorrow open, which 'I don't like it' does not.

ACROSS K-5

Older children lead the role-play, which is where the norm really gets set — a fifth-grader saying 'not for me today' and being met with a flat 'okay' teaches the room more than an adult explaining it. Give grades 4-5 the job of holding the no-commenting rule with each other, and be explicit that it applies to them first.

EXTEND IT Revisit the agreement at the halfway point and let the class amend it. A rule children have edited is a rule they enforce.

WATCH OUT The response to 'not for me today' is 'okay' and then a change of subject. Not 'are you sure', not a pause, not a look. Brief every adult — including parent volunteers and substitutes — in writing before they set foot in the room, because one well-meaning 'just one bite' undoes a term of this.

THE THREAD UNDERNEATH Saying what you need, and being met with an ordinary yes, is something children have to practice.

Listen to It First

Sound as a food property, and the easiest door in

Can you tell what someone is eating from the next room?

Grades K-2 & 3-5 40 min Light mess No heat needed Classroom is fine

YOU WILL NEED

- A range of sounds: crisps, raw carrot, apple, celery, bread, banana, grapes, a fizzy drink, dry cereal
- A screen or a sheet to hide the source
- A recorder or tablet
- The sound column of the word wall from card N01
- Headphones, offered to the whole group as a standing option

WHAT YOU DO

1. One child bites something behind a screen. Everyone else guesses what it was, and says what in the sound gave it away.
2. Build a loudness order: line up six foods from quietest to loudest and argue about the middle ones.
3. Record the same food bitten fresh and then bitten again after twenty minutes in the open air. Compare.
4. Pour cereal into a dry bowl and into a bowl of milk or water. Listen to what happens over two minutes.
5. Collect the words. Snap, crackle, squeak, thud, hiss, pop, crunch, crack, sizzle — add any the class invents.
6. Finish with a blind sound quiz between two teams.

EVERYONE PARTICIPATES

Nobody needs to eat anything: a child can be the screen-holder, the recorder, the scribe, the quizmaster or the guesser and never come within a meter of the food. The guessing game is the most fun anyone has in the first three weeks, and it is played entirely with ears.

WORTH ASKING

- What in the sound told you what it was?
- Same food, twenty minutes later, quieter. What changed inside it?
- Does loud mean fresh? Always?

JOURNAL Sound-words page, and the loudness order as a ranked list

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Crunch is a real, measurable food property and professional sensory panels test for it, because texture determines acceptance far more than flavor does. What a person hears when they bite something crisp is a rapid series of tiny structural failures — brittle cell walls fracturing all at once — and the sound reaches the ear through the jaw as well as through the air, which is why food sounds louder to the eater than to anyone else. That is a genuinely strange fact and children love it.

Crispness depends on the structure being dry and rigid. Let a crisp sit out and it picks up moisture from the air; the structure softens, the failures become gradual instead of sudden, and the sound goes. The same thing happens in fast-forward when cereal meets milk. So sound is a direct readout of moisture and structure, not decoration.

For this class, sound is also the strategically important sense. It is the only one that works at a distance. A child who is nowhere near touching a food can be the room's leading expert on how it sounds, and be right, and be seen to be right.

ACROSS K-5

K-1 are extremely good at this and often beat the older children, which is worth letting happen visibly. Grades 4-5 run the recording and the loudness ordering, and handle the argument about whether celery or carrot wins.

EXTEND IT Record and look at the waveforms on a tablet — a crisp bite is a tall spike, a soft one is a low mound, and children can match the picture to the sound. Grades 4-5: test whether a food that has gone quiet can be made loud again in a low oven, and explain the result.

WATCH OUT Screen for allergies before any food is bitten, and keep the biting behind the screen limited to children who have volunteered — never draft someone for it. Some children find mouth sounds genuinely distressing — that is misophonia and it is not squeamishness. Offer headphones to everybody at the start, not to anyone in particular, and make it easy to work at the recording-and-scribing end of the room.

THE THREAD UNDERNEATH There is more than one way to be the person who knows the most about something.

Three Cups, One Difference

The triangle test — how professionals ask whether a difference is real

You think these two taste different. How would you prove it?

Grades 3–5 50 min Light mess No heat needed Needs the kitchen

YOU WILL NEED

- Two versions of one thing, differing in exactly one way — lemonade at two sugar levels, or bread with and without salt
- Small identical cups, three per taster, coded with three-digit random numbers
- Water and plain crackers between samples
- Recording slips, and a tally chart
- A calculator for grades 4–5

WHAT YOU DO

1. Set the question: is there a difference people can actually detect, or do we only think there is?
2. Each taster gets three cups. Two are the same and one is different, in a random order they do not know.
3. Tasters take the three cups in the given order, water in between, and mark the odd one out. A taster has to pick one even with no idea — that is what makes the math work. Everyone else pours, codes, collects and tallies.
4. Collect the slips. Count how many people got it right.
5. Here is the math: with three cups, someone guessing at random gets it right one time in three. So what does your number mean?
6. Decide together whether the difference is real, and what would make you more confident.

EVERYONE PARTICIPATES

Tasting is optional here, exactly as everywhere else, and the roles that are not tasting are the ones that carry the session: the coder who assigns and records the random numbers is the only person who knows the answer and is therefore the most important person in the room. Also needed: the pourer, the order-randomizer, the slip-collector, the tallier and the statistician. Six jobs, one of which involves a mouth.

WORTH ASKING

- If nobody could taste anything at all, how many would still get it right?
- We got seven out of twelve. Are you convinced? What would convince you?
- Why did we use numbers instead of letters on the cups?

JOURNAL Triangle-test recording page — the result belongs to the class, not to a person

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

The triangle test is a standard method in professional sensory science, and running it properly teaches something most adults never learn: that *I can taste the difference* is a claim that can be tested, and that a person who is guessing will still be right sometimes.

The design features all matter: **Three-digit codes** instead of A/B/C, because A reads as first and best. **Randomized order**, because the first sample gets an advantage. **Forced choice** — you must pick one even if you have no idea — because that is what makes the one-in-three baseline exact. And **blind**, because expectation changes perception measurably; a taster who has been told one cup cost more reports differently, which is not a character flaw but a reproducible effect.

Then the arithmetic. Twelve tasters, four correct: that is exactly chance, and the honest conclusion is that this group could not detect the difference. Twelve tasters, ten correct: something real is going on. Somewhere in between is genuinely uncertain, and sitting in that uncertainty rather than resolving it is the most valuable ten minutes in the card. Professionals answer it with tables and probability; a fifth-grader can answer it with 'we should ask more people', which is the same instinct.

ACROSS K–5

Grades 4–5 design and run it; that is the real work. K–2 can be tasters if they wish and are better employed as the pouring and delivery crew, which requires care and secrecy and which they take extremely seriously. The one-in-three idea lands at about grade 3.

EXTEND IT Run it again with the labels showing and compare the two sets of results. The gap between them is the expectation effect, measured by the class. Grades 4–5: repeat with a smaller difference and find roughly where detection breaks down.

WATCH OUT Never report who got it right. The result is a class number and the slips go in the recycling in front of everyone. Screen allergies before pouring, and offer a non-tasting role to the whole group rather than to anyone in particular.

THE THREAD UNDERNEATH Being sure is not the same as being right, and there is a way to find out which one you are.

A Table Where Everyone Fits

Hosting, asking for what you need, and minding your own plate

What makes a person feel welcome at a table?

Grades K-2 & 3-5 55 min Real mess No heat needed Needs the kitchen

YOU WILL NEED

- Cloth, plates, cutlery, napkins, a jug and cups, something for the middle of the table
- Components served separately in bowls rather than pre-mixed
- Small 'not for me today' plates at every setting
- Cards with situations on them, for the role-play
- Something simple the class made earlier, to actually serve

WHAT YOU DO

1. Lay a table properly together. Talk about what each decision is for — the napkin, the jug within reach, the thing in the middle to look at.
2. Serve everything in separate bowls so each person builds their own plate. This is how the class always serves.
3. Practice three sentences out loud until they are ordinary: 'Not for me today.' 'Could you pass that?' 'Could I have mine without the sauce?'
4. Role-play the responses. When someone says 'not for me today', the answer is 'okay' and then a change of subject. Practice until it is boring.
5. Role-play the harder one: somebody comments on what someone else is eating. What do you say? What does the host say?
6. Eat, or not, together. The host's job is to notice whether everyone has what they need, not what anyone put on their plate.

EVERYONE PARTICIPATES

The host role is the one to notice here: it is high-status, it is about other people rather than about food, and it requires no eating. Rotate it every session. A child who hosts well has demonstrably participated fully in a meal, whatever was on their plate.

WORTH ASKING

- What does a good host actually do?
- Someone said 'not for me today'. What is the whole of your answer?
- Why would a rule about not commenting help everybody, not just one person?

JOURNAL The 'what I need at a table' page, and the table-plan sketch

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Two things are being taught and only one of them is manners. The first is **serving style**, which is a structural decision rather than a courtesy. Components served separately, with each person building their own plate, means nobody has to negotiate, nobody has to decline a dish in front of anyone, and mixed textures — the hardest thing for most sensory-driven eaters — are never imposed. Make it the standing way this class serves from the first week, so that it is the room's ordinary practice rather than an adjustment made for somebody.

The second is the **no-commenting norm**, and it needs to be taught explicitly rather than assumed, because peer policing is the real risk in a group. Children monitor and remark on each other's plates constantly unless given a rule and a reason. The rule is short — *in this kitchen nobody comments on what anybody else eats; your job is your own plate* — and it protects every child in the room, including the ones who eat unusual-looking food from home, the ones whose families eat differently for religious or cultural reasons, and the ones whose lunchboxes are thin some weeks.

Rehearsing the sentences until they are dull is the point. A script that has been said out loud twenty times in a calm room is available later in a room that is not calm.

ACROSS K-5

Pair a grade 4-5 host with a K-1 co-host and give the younger child a real duty — the jug, the napkins, the thing in the middle. Older children run the role-plays and should be asked to play the person who comments badly, then to say how it felt on the other side.

EXTEND IT Host the family night with the same rules posted at the door. Grades 4-5: write the class's own hosting guide for next year's group, in their own words.

WATCH OUT Adults are being watched constantly here. No adult comments on any child's plate, no adult comments on their own eating, and no adult says 'just try it' — including the parent helper who arrived ten minutes ago and means well.

THE THREAD UNDERNEATH Hospitality is noticing what someone needs without making them ask twice.

The Cucumber Weeps

Osmosis — water moving where you did not put it

You added salt, not water. So where did all that liquid come from?

Grades 3–5 50 min plus a 30-minute wait Real mess No heat needed Needs the kitchen

YOU WILL NEED

- Cucumbers, sliced thin, and a whole head of lettuce that has gone limp
- Salt, sugar, and plain water
- Clear bowls, a colander, a digital scale
- Raisins, and two glasses of water
- A timer and a recording sheet

WHAT YOU DO

1. Weigh 200 g of cucumber slices. Salt them heavily, put them in a colander over a bowl, and set the timer.
2. Weigh a second 200 g with no salt as the control. Same bowl, same room.
3. Put the limp lettuce in ice water. Put a second limp leaf in salted water.
4. Drop raisins into plain water and leave them.
5. At thirty minutes: weigh both cucumber batches again and measure the liquid collected. Do the arithmetic.
6. Check the lettuce and the raisins. One got crisp, one got limper, and the raisins got fat.

EVERYONE PARTICIPATES

Nothing here needs to be eaten. The jobs are weighing, timing, recording and calculating, and the numbers are the finding. A child who would never put a cucumber near their mouth can be the one who works out that 200 g became 142 g and announces it.

WORTH ASKING

- Nobody added liquid. Where did it come from?
- The lettuce got crisp in one bowl and floppier in the other. What is the rule?
- Why would salt keep food from going bad?

JOURNAL Observation page with a before-and-after weight table

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Every plant cell is wrapped in a membrane that lets water through easily and dissolved salt through only slowly. Water moves across such a membrane toward whichever side has more dissolved material — that is **osmosis**, and it needs no pump and no energy. Salt the outside of a cucumber slice and the outside suddenly has far more dissolved material than the inside, so water leaves the cells, the cells lose their internal pressure, and the slice goes floppy while a pool of liquid appears underneath.

That internal pressure has a name worth teaching — **turgor** — because it explains crispness. A crisp lettuce leaf is crisp because every cell is inflated and pressing against its neighbors. Put a limp leaf in plain water and osmosis runs the other way: water moves into the cells, pressure returns, and the leaf stiffens within twenty minutes. Put it in salt water and it collapses further. The raisins make the same point from the opposite end, swelling as water moves into their concentrated interiors.

This is not a curiosity. It is why salt is added to cucumbers before dressing them, why salted vegetables weep, and why a brine both seasons and changes texture. It is also the mechanism behind salt as a preservative — the same water-pulling effect that empties a cucumber cell empties a bacterial one.

ACROSS K–5

K–1 run the lettuce and the raisins, which are visual and dramatic, and squeeze the salted cucumber to feel the difference. Grades 2–3 do the weighing. Grades 4–5 compute percentage water loss and set up an extra condition of their own design.

EXTEND IT Run sugar against salt at the same weight and see which pulls more water. Grades 4–5: make quick pickles with the salted cucumbers and compare the texture against fresh by pressing and cutting — and by tasting, for anyone who wants to.

WATCH OUT Salted cucumber water is very salty and children will try to drink it. Say no in advance and tip it out where they can see.

THE THREAD UNDERNEATH Things move on their own, toward balance, whether or not anyone is pushing.

Pulling Colors Apart

Capillary action and paper chromatography

Is green one color, or is it several colors standing very close together?

Grades 3–5 55 min Real mess No heat needed Needs the kitchen

YOU WILL NEED

- Spinach leaves, and a mortar or the back of a spoon
- Coffee filters or chromatography paper, cut into strips
- Rubbing alcohol (adult handled) for the spinach, and water for the food-coloring run
- Black and green water-based marker pens, and liquid food coloring
- Tall narrow glasses, pencils and tape to suspend the strips
- Celery stalks with leaves, and colored water

WHAT YOU DO

1. Start the celery: split one stalk lengthways halfway up, stand each half in a different color of water, leave it for the session.
2. Draw a fat dot of black marker 2 cm from the end of a paper strip. Hang it so the end touches water but the dot does not.
3. Watch. The water climbs, reaches the dot, and carries the inks up the paper at different speeds.
4. For spinach: crush the leaves, add a little alcohol to extract the pigment, dab the extract onto a strip, and run it in alcohol.
5. Measure how far each color band traveled and how far the solvent front traveled. Record both.
6. At the end of the session, cut the celery across and look at where the color went.

EVERYONE PARTICIPATES

No food contact required at all. This is a card about paper, water and color that happens to use vegetables as the source. Jobs: paper-cutter, strip-hanger, timer, measurer, and the person who draws the finished chromatogram into the journal.

WORTH ASKING

- Black split into how many colors? Were they hiding, or were they made?
- Why did some colors go further than others?
- What does the celery tell us about how a plant drinks?

JOURNAL Tape the actual chromatogram into the journal — it is the data

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Two ideas, one demonstration. **Capillary action** is water climbing a narrow space without being pushed: water molecules stick to the paper fibers (adhesion) and to each other (cohesion), so the leading edge drags the rest along. The narrower the channel, the higher it climbs. The celery is the same effect in a living system — its xylem vessels are narrow tubes, and a split stalk in two colors produces a two-colored leaf, which is a startling thing to watch happen in one afternoon.

Chromatography rides on that. As the solvent climbs it carries dissolved pigments with it, and each pigment travels at its own speed depending on how much it prefers dissolving in the moving solvent versus sticking to the paper. Pigments that prefer the solvent race ahead; pigments that cling stay low. Black ink separates into an unexpected set of colors, which is the moment children stop thinking of black as a color and start thinking of it as a mixture.

The spinach run is the real payoff. Green leaves contain chlorophylls and also carotenoids — the yellows and oranges — which were there the whole time, masked. It is the same reason leaves turn color in autumn: nothing is added, the green is simply withdrawn.

ACROSS K–5

K–1 run the marker-and-water version, which works perfectly and is beautiful. Grades 2–3 measure band distances. Grades 4–5 do the spinach extraction with adult-handled alcohol and compute the ratio of pigment distance to solvent distance, which is how chemists actually report it.

EXTEND IT Compare green leaves with red or purple ones and connect the result to card S06. Grades 4–5: run the same extract in water and in alcohol and explain why the results differ.

WATCH OUT Rubbing alcohol is flammable, an irritant and a poison. An adult does every pour and holds every open container; children never handle it. No ignition source anywhere in the room — gas burners and pilot lights off, no hot plate, no oven in use; if the kitchen runs on gas, do this card in another room. A few milliliters only, in a covered tall glass away from where children lean, windows open. Glassware is dedicated and does not go back into food service.

THE THREAD UNDERNEATH What looks like one thing is often several things you have not separated yet.

Three Ways to Set a Liquid

Gelatin, agar and pectin — the gels do not behave alike

All three turned juice into something solid. Are they the same trick?

Grades 3–5 60 min plus setting time Real mess Adult at the heat Needs the kitchen

YOU WILL NEED

- Unflavored gelatin, agar powder, and powdered pectin
- Fruit juice, sugar, and lemon juice
- A saucepan, a thermometer, small clear cups and a slotted spoon
- A cold plate straight from the freezer, for the jam wrinkle test
- A warm water bath, for the melting comparison

WHAT YOU DO

1. Set three identical juice samples: one with gelatin, one with agar, one with pectin plus sugar and lemon.
2. Record the temperature at which each one sets as it cools.
3. Once all three are firm, put one cup of each into a warm water bath at about body temperature. Watch for ten minutes.
4. Cut a cube of each and press it. Describe the difference — one shatters, one wobbles, one smears.
5. Do the jam test properly: spoon a drop onto the frozen plate, wait thirty seconds, push it with a finger. A set drop wrinkles.
6. Add a spoon of fresh pineapple to a fourth gelatin cup and a fourth agar cup. Predict what happens to each.

EVERYONE PARTICIPATES

Setting, measuring, timing and testing are the work; eating is not part of the card. The frozen-plate wrinkle test is a genuine professional technique and it requires a finger and a cold plate, nothing else.

WORTH ASKING

- One melted in the warm bath and one did not. What does that tell you about what they are made of?
- Why does a jam recipe insist on the lemon?
- Predict the pineapple cups. Now explain your prediction.

JOURNAL Three-column comparison table, plus a drawing of each cut cube

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

A gel is a liquid trapped inside a three-dimensional mesh, and the three setting agents build that mesh out of completely different materials — which is why they behave so differently and why substituting one for another rarely works.

Gelatin is protein. Its mesh melts at around body temperature, which is precisely why gelatin desserts feel as though they dissolve in the mouth — they literally do. Like agar it is thermoreversible: melt it and cool it and it sets again — the pectin gel is the odd one out, since once set it does not re-melt. **Agar** is a polysaccharide from seaweed. It sets at a higher temperature, holds firm at room temperature and well above, and does not melt in the mouth, so it gives a cleaner, more brittle bite — it is also what microbiologists grow cultures on, for exactly that heat tolerance. **Pectin** is a polysaccharide from fruit, and it is the fussiest: the standard kind needs both acid and a high concentration of sugar to form its network at all, which is why jam recipes are so insistent about the lemon and the sugar quantities.

The pineapple test is the elegant part. Fresh pineapple carries an enzyme that cuts protein chains, so it destroys a gelatin mesh and leaves the cup liquid — but agar is not a protein, so a protein-cutting enzyme has nothing to cut and the agar sets regardless. One experiment, and the children can tell you what each gel is made of.

ACROSS K–5

Grades 4–5 handle the thermometer and the melting comparison. K–2 do the pressing and describing, which feeds straight into the texture vocabulary wall, and run the wrinkle test. Everyone predicts before the pineapple goes in.

EXTEND IT An adult makes a fruit jam start to finish with no child at the pan — boiling jam spits at over 100°C — and children run the frozen-plate test on a cooled spoonful. Grades 4–5: find the minimum amount of each setting agent that will still set a fixed volume of juice, and report it as a percentage.

WATCH OUT Hot sugar mixtures for jam are among the hottest things in a kitchen and are adult-only, start to finish. Agar must be boiled to activate and the pan stays with an adult. Gelatin is animal-derived, usually pork or beef. Ask on the intake form about animal-derived materials and not only foods, and set up a fish-gelatin or agar-only version of every station as the standing arrangement rather than as somebody's alternative.

THE THREAD UNDERNEATH Two things that look the same from outside can be built completely differently.

The Apple Goes Brown

Enzymatic browning, and four ways to stop it

Nothing was added and nothing was cooked. So why is it brown?

Grades K-2 & 3-5 45 min plus checking back Light mess Adult at the heat Needs the kitchen

YOU WILL NEED

- Apples, pears, bananas, avocado and potato — all brown, at different speeds
- Lemon juice, plain water, salted water, plastic wrap, a bowl of ice
- A pan of boiling water and a slotted spoon, for the blanching condition
- Labeled plates and a camera
- A timer

WHAT YOU DO

1. Cut one apple into six equal pieces, fast, so they all start at the same moment.
2. Piece 1: left in the air, the control. Piece 2: lemon juice. Piece 3: under water. Piece 4: wrapped airtight.
3. Piece 5: dropped in boiling water for sixty seconds, then cooled. Piece 6: straight into the ice bowl.
4. Photograph the whole set at 0, 15, 30 and 60 minutes.
5. Rank them from least to most browned. Then work out what each treatment actually removed or changed.
6. Repeat with a banana and an avocado to see whether the same ranking holds.

EVERYONE PARTICIPATES

Cutting, timing, photographing, ranking. No tasting is involved in the experiment at all, and the browned apples are data at the end rather than snacks — say so at the start so nobody is waiting to find out.

WORTH ASKING

- Which treatment took away the oxygen? Which one broke the enzyme?
- The cold one went brown eventually. What does that tell us about cold?
- Why does a cut potato go brown but a whole one does not?

JOURNAL Photo strip or drawing at four time points, with the ranking underneath

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Cutting an apple breaks cells open and lets two things meet that were previously kept apart: an enzyme called polyphenol oxidase and the phenolic compounds stored in the fruit. With oxygen from the air, the enzyme converts those compounds into products that link up into brown pigments. Nothing was added and nothing rotted — the fruit simply did this to itself, and it is the same chemistry that makes tea leaves and cocoa beans brown deliberately.

Because three ingredients are required — enzyme, phenolics, oxygen — the six conditions sort onto four ways of interfering. **Remove the oxygen:** water and plastic wrap both work by keeping air off the surface. **Change the pH:** acid from lemon juice drops it below the range the enzyme works well in. **Destroy the enzyme:** sixty seconds in boiling water denatures it outright and permanently. **Slow everything down:** cold does not stop the reaction, it just reduces the rate, so the iced piece browns eventually.

Getting children to sort the treatments by *which ingredient it removed* rather than by how well it worked is the whole lesson, and it turns a list of kitchen tips into one mechanism.

ACROSS K-5

K-1 do the ranking and the photography, and they are good at spotting small color differences. Grades 2-3 run the timer and the labeling. Grades 4-5 handle the blanching with an adult and take on the sorting-by-mechanism question, which is the hard part.

EXTEND IT Test whether other acids work as well as lemon — vinegar, orange juice, pineapple. Grades 4-5: time how long each fruit takes to reach the same browning level and explain the differences in terms of how much enzyme each contains.

WATCH OUT Blanching is adult-only. Avocado pits and skins are slippery — an adult cuts and removes them.

THE THREAD UNDERNEATH Three things had to meet for this to happen, and there are four ways to break up the meeting.

Your Tongue Is Not My Tongue

How plants defend themselves, and why people taste them differently

Two people eat the same broccoli. Why does one of them find it bitter?

Grades 3–5 55 min Real mess No heat needed Needs the kitchen

YOU WILL NEED

- Onions, a chilled one and a room-temperature one, plus a sharp knife and a blunt one
- Raw broccoli, kale or Brussels sprout, in small pieces
- Mild and hot chilli, or a drop of hot sauce, for the optional capsaicin part
- Milk, water, and something fatty, for the capsaicin test
- Swimming goggles, which are funny and which genuinely work

WHAT YOU DO

- Cut a room-temperature onion with a blunt knife. Time how long before the first person's eyes sting.
- Cut a chilled onion with a sharp knife. Time it again. Change one thing at a time and note which change helped most.
- Try goggles. Try cutting near a running faucet. Collect everyone's family remedy and test the ones that are safe.
- Everybody who wants to tries a small piece of raw brassica and rates the bitterness from 1 to 5 — privately, on a slip.
- Collect a slip from everyone, including a 'didn't taste' box. Put the bitterness numbers on the board with no names — and say the number of slips returned, never the number of children who tasted.
- Talk about what that spread means. Nobody is wrong. The tongues are genuinely different.

EVERYONE PARTICIPATES

The tasting itself is visible — that cannot be helped in a room. What the slips protect is the rating, not the fact of tasting. So make the slip universal: everyone submits one, and the slip carries a 'didn't taste' box, so the stack that comes back is the same size as the class. The onion half needs no eating at all and is the funniest ten minutes of the term.

WORTH ASKING

- Which change helped most — cold, sharp, or water?
- Our numbers went from 1 to 5 for the same piece of broccoli. How can that be?
- If two people taste the same thing differently, is one of them wrong?

JOURNAL The anonymous class spread on a bar chart, plus 'what I taste that others might not'

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Plants cannot run, so many of them make chemicals instead, and several of the strongest flavors in a kitchen are defenses. Cut an onion and you rupture cells, letting an enzyme reach compounds it is normally kept away from; the reaction produces a volatile sulfur compound that drifts up, reacts with the water in the eye, and stings. Everything that helps is a way of interfering with that: chilling slows the enzyme, a sharp blade ruptures fewer cells than a blunt one crushing its way through, and water carries the compound away before it reaches a face.

The bitterness half is the important one. Brassicas contain bitter compounds, and human beings vary — substantially and heritably — in how strongly they taste them. Some people carry receptor variants that make certain bitter compounds intensely unpleasant; others barely register the same molecule. This is a well-documented genetic difference, not a matter of willpower or fussiness.

That is worth teaching plainly to a whole class, because it makes something true and useful common knowledge: **when two people disagree about how something tastes, they can both be reporting accurately.** A child who has heard that from a teacher, in front of everyone, in a lesson about plants, has been handed a defense they can use for years — and no child had to be singled out for it to be delivered.

ACROSS K–5

Grades 4–5 run the onion variables and the anonymous tally. K–2 are the timekeepers and the goggle-wearers. Everyone hears the conclusion, and it matters to anyone who has ever been told they are being difficult about food — nobody has to identify themselves for it to land.

EXTEND IT Test whether cooking changes the bitterness rating, and re-run the anonymous vote. Grades 4–5: research why milk helps with chilli heat and water does not — the compound barely dissolves in water but dissolves readily in fat — and design a test.

WATCH OUT Chilli is optional, mild only, tiny amounts, and never near eyes; hands washed immediately afterward, and have milk on hand. Adults cut all onions for K–2. Never press a child to taste a bitter sample — the anonymity only protects them if the tasting is genuinely free.

THE THREAD UNDERNEATH 'It tastes different to me' is a true sentence, and now everyone in this room knows why.

Four Families of Color

What heat and acid do to the pigments in plants

Why do green beans go drab in the pan, and beets go everywhere?

Grades 3–5 55 min Real mess Adult at the heat Needs the kitchen

YOU WILL NEED

- Green beans or broccoli (chlorophyll); carrot or sweet potato (carotenoid)
- Red cabbage or blueberries (anthocyanin); beet (betalain)
- Vinegar, baking soda, water, and a little oil
- Small pans or heatproof bowls, a hot plate, a timer
- White plates for comparison, and a camera

WHAT YOU DO

1. Set up four columns, one per pigment family. Photograph every raw sample first.
2. Cook a sample of each in plain water for two minutes, and a second sample for ten. Compare.
3. Cook a third sample of each in water with a splash of vinegar. Compare again.
4. Cook a fourth in water with a pinch of baking soda. The green one will go vivid and the texture will go to mush.
5. Separately, warm carrot pieces in a little oil and look at the oil after five minutes.
6. Build the summary table: which pigments moved into the water, which changed color, which did neither.

EVERYONE PARTICIPATES

This is a looking-and-recording card. Photographing, timing, building the table and writing the summary are the work. The cooked samples are evidence and are treated as such.

WORTH ASKING

- Which colors ran into the water? What do those have in common?
- The baking soda one is the greenest and the mushiest. Would you use it?
- Predict what red cabbage will do in vinegar, before we try it.

JOURNAL Four-by-four color table, painted or colored in

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Four pigment families, four sets of rules, and once a child has the table they can predict a great deal of what a pan will do.

Chlorophyll is green, fat-soluble, and fragile in acid — acid displaces the magnesium at the center of the molecule and the color shifts to a dull olive, which is exactly what happens to green vegetables left too long in a pan with anything acidic. Alkali does the reverse and locks the green in brilliantly, but it also attacks the cell walls, so the vivid sample is mushy: a real trade-off children can see and taste-free judge. **Carotenoids** are the oranges and yellows, fat-soluble and remarkably stable to both heat and acid — which is why the carrot stays orange and why the oil turns orange instead. **Anthocyanins** are water-soluble and pH-sensitive, so they leach into the cooking water and change hue with acid and alkali. **Betalains**, the beet pigments, are water-soluble and unstable to prolonged heat, which is why beets bleed into everything and fade if overcooked.

The generalization worth landing: a pigment that dissolves in water leaves the vegetable, and a pigment that dissolves in fat stays put or moves into the oil. Solubility predicts behavior.

ACROSS K–5

K–1 sort the raw and cooked samples by color and take the photographs. Grades 2–3 run the timers. Grades 4–5 build the four-by-four table and write the predictive rule, then test the rule on a vegetable nobody has tried yet.

EXTEND IT Connect straight to the dye card A01: the pigments that leach are exactly the ones that make good dyes. Grades 4–5: work out why chefs blanch green vegetables briefly and then plunge them into ice water.

WATCH OUT Hot pans and boiling water stay with adults. Baking-soda-cooked vegetables are slippery and taste strongly of soda — they are a demonstration, not lunch.

THE THREAD UNDERNEATH One rule — what dissolves in what — explains four different things at once. That is what a good rule does.

Closer Than You Have Ever Looked

Magnification, crystals and the structure of ordinary things

Salt and sugar look identical in a bowl. Do they look identical up close?

Grades K-2 & 3-5 50 min Light mess No heat needed Classroom is fine

YOU WILL NEED

- Hand lenses, and a clip-on macro lens for a tablet if you can get one
- Salt (table and coarse), sugar (granulated and powdered), flour, cornmeal, semolina, poppy seeds
- Dark paper and white paper, to look against
- Slices of potato and apple, plus a drop of iodine if your school permits it
- Drawing paper and sharp pencils

WHAT YOU DO

1. Look at salt and sugar side by side under the lens, on dark paper. Draw what you see, large.
2. Predict which is which from the drawings alone, then check.
3. Work through the other powders. Sort them by particle shape, not by name.
4. Photograph through the lens with a tablet if you have one, and print the best images.
5. Look at the cut face of a potato and an apple. Look for structure, not color.
6. Build a 'known samples' page: each powder drawn and labeled, to be used later to identify an unknown.

EVERYONE PARTICIPATES

A looking card, start to finish, and one of the best for a child who wants a quiet table and no food contact. Nothing is tasted, nothing is cooked, and the deliverable is a drawing. It also pairs naturally with card A04, so a child who likes drawing has two strong sessions in the term.

WORTH ASKING

- Which one is salt? How do you know, without tasting it?
- Powdered sugar and granulated sugar are the same substance. Why do they behave so differently?
- Does what you saw explain how it feels in your mouth?

JOURNAL Known-samples drawing page — used again later in the course

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Salt crystals are cubes, and they are cubes because of how the sodium and chloride ions stack — the shape at the scale you can see is a direct consequence of the arrangement at a scale you cannot. Table sugar crystals are not cubes; they are slanted blocks with a different symmetry entirely. Once a child has drawn both, they can identify an unlabelled sample from across the room, which is a genuinely satisfying superpower and also the actual logic of mineralogy.

The rest of the powders sort into their own categories: flour is irregular and dusty, semolina is angular and uniform, cornmeal is chunky, powdered sugar is sugar that has been ground until barely a crystal face survives, which is why it dissolves so fast — there is far more surface for water to reach. It also carries a little cornstarch to stop it caking, which children will spot under the lens and ask about.

Magnification is also a door into the sensory work. Grittiness, smoothness and 'powderiness' are mouthfeel properties children struggle to explain, and seeing the particles makes the words land: semolina feels different from flour because the particles genuinely are different, and the child has now seen it rather than been told.

ACROSS K-5

K-1 sort and draw shapes. Grades 2-3 build the known-samples page. Grades 4-5 run the tablet macro photography and take on the unknown-identification challenge, which they should design for the younger children.

EXTEND IT Grow salt and sugar crystals on a string over a week and look at them under the lens as they grow. Grades 4-5: if iodine is permitted, stain potato and observe starch grains, and compare potato starch with rice starch.

WATCH OUT Iodine stains skin and clothing and is not universally permitted in schools — check first, and if in doubt leave it out; the card works fully without it. Fine powders should not be blown or inhaled.

THE THREAD UNDERNEATH Ordinary things are strange if you get close enough. That is available to anyone with a lens.

The Cracker Turns Sweet

Digestion as a system — mechanical and chemical

Chew a plain cracker for two minutes without swallowing. What happens to the taste?

Grades 3–5 45 min Light mess No heat needed Classroom is fine

YOU WILL NEED

- Plain unsalted crackers, or plain cooked rice
- A timer
- A long piece of rope or a marked strip of floor, for the digestive-tract scale model
- A zip bag, a little water and a cracker, for the stomach model
- Paper and pens for the system diagram

WHAT YOU DO

1. An adult, and anyone else who wants to, chews a plain cracker without swallowing for two minutes. Everyone else times it and writes down what the chewers report — reporting what somebody else noticed is the same job.
2. Ask what changed. It gets sweeter. Nothing sweet was added.
3. Build the mechanical model: cracker and water in a zip bag, squeezed by hand. Watch it break down without any chemistry.
4. Lay out the rope model of the digestive tract to rough scale and walk its length.
5. Draw the system as a diagram with inputs, stages and outputs, using the words mechanical and chemical.
6. Name where each thing happened: teeth, saliva, stomach, small intestine.

EVERYONE PARTICIPATES

Chewing the cracker is optional and plain crackers are a commonly accepted food, which is deliberate — this is one of the few cards where a taste is central, so the item is chosen to be as low-demand as possible. A child who does not chew one runs the rope model, the zip-bag model and the diagram, and can time everyone else.

WORTH ASKING

- The chewers said it got sweeter. Nothing sweet was added — where did it come from?
- Which part of this is machinery and which part is chemistry?
- Where else in this course have we met an enzyme?

JOURNAL System diagram page

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Saliva contains an enzyme that cuts long starch chains into much shorter ones, and the sugar it leaves behind — maltose — is sweet. A cracker is mostly starch, which is tasteless; chew it long enough and the enzyme has time to work, and the sweetness that appears is real sugar that was not there when the cracker went in. Breaking that sugar down the rest of the way happens later, in the small intestine. It is the clearest, cheapest demonstration of enzyme action available.

That gives you the frame for the whole system: digestion is **mechanical** — teeth, the stomach's churning, the gut's squeezing — and **chemical** — enzymes and acid taking molecules apart. Both are needed, and both start earlier than children expect. The zip-bag model isolates the mechanical half so they can see how much breakdown happens with no chemistry at all.

A note on how to teach this: keep it structural. This card is about how a system works, in the same way a card about a pump or a filter would be. It is not about which foods are good or bad, how much anyone should eat, or what food does to a body's size, and any drift in that direction should be steered straight back to the mechanism. The interesting question is *how does the cracker get taken apart*, and there is plenty in that to fill a session.

ACROSS K–5

K–1 walk the rope and squash the bag. Grades 2–3 label the stages. Grades 4–5 draw the system diagram properly with inputs and outputs and explain the enzyme step to the younger children, which is the real test of whether they have it.

EXTEND IT Grades 4–5: test whether a cracker chewed and then placed in water behaves differently from one simply soaked — a test with no mouths in it at all. Then diagram where in the system each stage of breakdown happens.

WATCH OUT Screen for allergies, use one specific cracker, and name it in advance to families. Keep the whole session away from anything about amounts eaten, body size, or foods being good or bad — if a child raises it, acknowledge it briefly and return to the mechanism.

THE THREAD UNDERNEATH Your body is doing something complicated right now without asking you. That is worth knowing about.

The Color in the Peel

Natural dyes, mordants and colorfastness

Onion skins go in the trash. What if they went on a shirt instead?

Grades K-2 & 3-5 90 min over two sessions Cover the table Adult at the heat Needs the kitchen

YOU WILL NEED

- Dyestuffs: yellow onion skins, turmeric, red cabbage, beets, avocado pits, black beans
- White cotton squares — old sheets cut up — plus watercolor paper
- Alum (spice aisle) and salt for mordanting; strong black tea for the cotton tannin bath; vinegar and washing soda for shifting color
- Stainless pots, a hot plate, tongs, jars, rubber bands and string
- A clothesline, and a sunny windowsill for the fade test

WHAT YOU DO

1. Session one: simmer each dyestuff in its own pot for 30–45 minutes, then strain. That is a dye bath.
2. Mordant the cotton first — a bath of strong cooled black tea, then a simmer with alum and salt, rinsing between. Leave one square unmordanted as the control.
3. Tie some squares with string or rubber bands. Where the string is tight, the dye cannot reach.
4. Steep. Ten minutes gives a pale shade, overnight a deep one. Record how long each piece stayed in.
5. Rinse, dry, and lay the results out as a labeled color library.
6. Session two: dip one corner of each cabbage-dyed piece in vinegar and another in washing-soda water. Then tape half of every square under card on a sunny sill for two weeks.

EVERYONE PARTICIPATES

Nobody eats anything on this card, which makes it a good early session — a child still deciding whether this kitchen is safe can have their hands in food for two hours with no possibility of a taste being expected. Jobs: simmer-watcher, tier, timer, rinser, color-librarian, label-writer. The pH-shifting job at the end is the most dramatic in the session and needs no contact with food at all.

WORTH ASKING

- The two squares went in the same pot. Why did only one keep its color?
- Which of these would you trust on a shirt you actually wear? What is your evidence?
- The string stopped the dye. What else could we use to block it?

JOURNAL Color library page, and the two-week fade comparison on a graph page

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Dyeing is chemistry that happens to be beautiful, and the mordant is the part that makes it art rather than a stain. Alum works by leaving metal ions bound to the fiber, so the dye molecule bonds to the metal rather than merely sitting on the surface. It binds readily to protein fibers — wool and silk — and poorly to cotton, which is why cotton gets a tannin bath first and then the alum. Expect the effect on cotton to be real but modest, and expect turmeric to defy it: turmeric is a direct dye that stains cotton perfectly well with no mordant at all, which is an exception worth showing rather than hiding.

The plants behave differently. Turmeric is brilliant and notoriously fugitive — it fades fast in sunlight, which the windowsill test proves rather than asserts. Onion skin gives a warm gold that is strong in the pot and disappointing on the sill; it is a classic fugitive dye too. Red cabbage anthocyanin is the pH-sensitive one, so vinegar swings it pink and washing soda blue-green, and a child can move a finished piece along the color scale after it is dry. Beets fade quickly, and saying so in advance turns a letdown into a finding.

The tied resist is the oldest patterning technology there is, appearing independently in shibori, bandhani and tie-dye. The rule is physical, not decorative: dye reaches what water reaches.

ACROSS K-5

Tying is the fine-motor bottleneck, so pair a grade 4–5 tier with a K–1 holder and let the younger child choose where the ties go. Older children manage the pots and the timing log; younger children manage the drying line and the labeling, which is a real job because a color library with no labels is useless.

EXTEND IT Dye paper and bind it into the class cookbook covers. Grades 4–5: make an iron mordant by rusting steel wool in vinegar for a week, and discover that the same dye bath gives completely different colors over iron than over alum — the mordant, not the plant, is doing half the work.

WATCH OUT Every pot, tong and jar used for dyeing is dedicated, labeled and stored away from food equipment, permanently. Alum is a powder and an eye and airway irritant: an adult weighs and adds it, with children away from the bench, and the mordant pot is adult-run throughout. Gloves for wet mordanted cloth; pots stay at the back and are never carried by a child; baths are cooled before straining. An adult halves the avocado pits. The iron-mordant extension is adult-only — the jar gives off hydrogen and must never be sealed, and steel wool is a fire hazard.

THE THREAD UNDERNEATH Something everyone throws away turned out to be worth keeping. That noticing is a habit, not a fact.

A Color Wheel You Can Eat

Primary, secondary and complementary color, in produce

Can you build a whole color wheel without opening a paint box?

Grades K-2 & 3-5 50 min Light mess No heat needed Classroom is fine

YOU WILL NEED

- A wide produce spread: lemon, carrot, tomato, radicchio, red cabbage, blueberry, kiwi, spinach, yellow pepper
- A large paper circle, pre-divided into twelve segments
- Paper plates, and one small plate per child for their own composition
- Colored pencils or watercolor, for recording
- A phone or tablet to photograph the finished wheel from directly overhead

WHAT YOU DO

1. Find the three primaries in food first. Red and yellow are easy. Blue is genuinely hard — send a team hunting.
2. Place each food on the wheel segment it belongs to. Argue about the ones that sit between two.
3. Photograph the completed wheel from overhead.
4. Now find the opposites: what sits directly across from the carrot? From the radicchio?
5. Each child builds a small plate using one complementary pair only — two colors, nothing else.
6. Then a second plate in a single color family, light to dark. Compare which plate people's eyes go to first.

EVERYONE PARTICIPATES

Color is the one sensory channel every child in the room has full access to regardless of where they are with food, which is why this card is worth running in week one. Nothing is tasted. The task is sorting, arguing and arranging, and the child who never intends to eat a radicchio can be the person who correctly places it.

WORTH ASKING

- Where is the blue? Why do you think there is so little of it?
- These two plates have the same food on them. Why does one look louder?
- If you wanted someone to notice one thing on a plate, what color would you put behind it?

JOURNAL Color wheel page, plus the plate-design page

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Color is the fastest thing a person judges food by — faster than smell, far faster than taste — so it is worth teaching deliberately rather than leaving to chance. The wheel does two jobs here. It teaches the structure (primaries mix to secondaries; opposites across the wheel intensify each other) and it makes an honest problem visible: there is almost no true blue in food. Blueberries are violet, red cabbage is purple, blue corn is gray-violet. Children usually find this frustrating and then find it interesting, and it is a real observation about pigment chemistry — the anthocyanins that come closest are pH-sensitive and rarely land on blue.

The complementary-pair plate is the design lesson. Orange carrot against deep blue-violet cabbage reads as more intense than either does alone, because the eye judges a color partly against what surrounds it. The monochrome plate teaches the opposite move — restraint, and value (light to dark) doing the work that hue usually does. Both are compositional choices a child can then make on purpose for the rest of the course.

ACROSS K-5

Kindergarteners sort by color, which they are good at and enjoy. Grades 2-3 handle the secondaries and the arguments about in-between cases. Grades 4-5 run the complementary-pair analysis and the overhead photograph, which needs a steady hand and a decision about framing.

EXTEND IT Photograph both plates and convert the images to grayscale. The complementary plate often collapses into sameness while the monochrome plate stays legible — which is a real lesson about value versus hue, and a surprise every time.

WATCH OUT Beets and turmeric stain hands, clothes and laminate. Aprons on, and put a tray under everything.

THE THREAD UNDERNEATH You can arrange something so that other people see what you want them to see. That is a kind of power.

The Okra Is a Stamp

Relief printmaking, repeat pattern and the edition

Cut a vegetable in half. What did you just make?

Grades K-2 & 3-5 70 min Cover the table No heat needed Classroom is fine

YOU WILL NEED

- Okra, celery hearts, lotus root, star fruit, bell pepper, mushroom, corn on the cob, cabbage wedges
- Potatoes, for carved stamps, and lino tools or plastic clay tools for grades 4-5
- Block printing ink or thick tempera, a brayer, and a flat tray to roll out on
- Newsprint for proofs, then good paper or plain cotton tote bags for the edition
- Paper towels, and a marked drying rack

WHAT YOU DO

- Cut a range of produce across the grain and look at the cut faces before inking anything. Some are already patterns.
- Roll a thin even film of ink onto the tray. Press the cut face in. Too much ink turns the print to mush — prove it once on purpose.
- Pull test prints of every vegetable on newsprint. Label each one with what made it.
- Choose one motif and print it as a repeat: a grid, a half-drop, a rotation. Mark registration guides in pencil first.
- Grades 4-5: carve a potato so that the raised part prints and the cut-away part does not. Predict what will happen to anything you carve as a letter.
- Print an edition — six copies of the same design, as close to identical as you can make them. Number them.

EVERYONE PARTICIPATES

No tasting, no food contact for anyone who does not want it — the vegetables are tools here, not food, and saying that out loud at the start changes the room. Jobs: cutter, inker, printer, puller, registration-marker, drying-rack manager. The inker has the most technical job in the session.

WORTH ASKING

- What is inside this vegetable that we could not see before we cut it?
- Your first print was crisp and the fourth is blurry. What changed?
- You carved your name and it came out backwards. Now what?

JOURNAL Pattern page — print directly into the journal

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Relief printing is one idea: the high parts print, the low parts do not. Every cut vegetable is already a relief block, made by the plant rather than by a person, and the cut face reveals structure that is otherwise invisible — okra's five-pointed star is its seed chambers, celery's crescent is the shape of the leaf stalk itself, with its vascular bundles printing as a row of dots inside the curve, lotus root's rings are air channels for a plant that grows in mud. So the print is simultaneously a piece of design and a botanical cross-section, which is why this card sits next to the illustration card.

Two pieces of real printmaking craft are worth insisting on. The first is ink thickness: a thin even film gives a crisp print and a thick one fills the detail, and children only believe this after ruining one. The second is that a carved block prints in mirror image, which matters the instant a child carves a letter or a name. Let them discover it rather than warning them, then let them solve it.

An **edition** — a set of prints from one block, numbered — introduces a genuinely different idea about art: that an original can exist in multiples, and that consistency is itself a skill.

ACROSS K-5

K-1 print freely with pre-cut produce and make the wrapping paper. Grades 2-3 run repeat patterns with registration guides. Grades 4-5 carve blocks, discover mirror-writing, and print the editions. Older children cut all produce; nobody under grade 4 uses a carving tool.

EXTEND IT Print onto the tote bags families take home, or onto paper that becomes the cookbook endpapers. Grades 4-5: two-color printing with a second block and registration marks, which is genuinely hard and worth the frustration.

WATCH OUT Adults cut all produce; wet vegetables skid badly. Use water-based inks only, and check them for latex and for any food allergens in the binder. Inked vegetables are no longer food — they go in the compost, not a mouth. If lino tools are used at all: bench hook mandatory, cut away from the body, the holding hand never in front of the blade, one adult supervising that table and nothing else. Plastic clay tools or pre-carved blocks lose nothing. Specify certified non-toxic water-based ink. Inked produce goes in the trash, not the compost. Star fruit is a specimen here and is never eaten: carambola is dangerous for anyone with impaired kidney function.

THE THREAD UNDERNEATH Making six of something that match is a different skill from making one good one.

Draw It Before You Know It

Observational and botanical illustration

How long can you look at one pepper before you start drawing?

Grades K-2 & 3-5 60 min Light mess No heat needed Classroom is fine

YOU WILL NEED

- One specimen per child: pepper, pear, garlic bulb, artichoke, pomegranate, fennel, romanesco
- Graphite pencils in two grades, kneaded eraser, and watercolor with one round brush
- Heavy paper, and a magnifier or a phone macro lens
- A ruler, for the scale bar
- A sharp knife, adult-operated, for the cross-sections

WHAT YOU DO

1. Look first and draw nothing for two full minutes. Time it — it feels much longer than it is.
2. Contour drawing: draw the outline without lifting the pencil and without looking at the paper. The results are funny and they fix how children look.
3. Now the real drawing. Whole specimen first, drawn large enough to fill the page.
4. Adult cuts each specimen in half. Draw the inside. This is usually the better drawing.
5. Add the three scientific conventions: a scale bar, label lines that do not cross, and the specimen's name.
6. Lay a light watercolor wash over the graphite, working from light to dark and leaving the highlights as bare paper.

EVERYONE PARTICIPATES

This card is entirely about looking, so it is fully accessible to every child in the room with no negotiation. Children who find the kitchen loud or smelly can work here at a quiet table with the same assignment as everyone else. A child who would not put a pomegranate near their mouth may produce the best drawing of one in the class, and that is a real and visible success in front of peers.

WORTH ASKING

- What did you notice in minute two that you had not noticed in minute one?
- Is your drawing of this pepper, or of peppers in general? Which is more useful, and to whom?
- Why would a scientist want a drawing when a camera exists?

JOURNAL Full-page specimen frame, and the cross-section page

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Botanical illustration exists because a drawing can do something a photograph cannot: it can show what is typical rather than what happened to be in front of the lens, and it can hold a cutaway, a detail and the whole specimen on one sheet at a chosen scale. That is why scientific illustrators are still employed, and it is worth telling children, because it reframes drawing from decoration into a method for seeing.

The two-minute silent look is not a mindfulness exercise, it is the technique. Children draw the pepper they have in their head — a symbol, four lobes, a green stalk — unless something stops them long enough to see the actual asymmetry in front of them. Contour drawing interrupts the same habit from the other side by making the symbol impossible to execute.

The conventions matter too. A **scale bar** tells a future reader the size without depending on the reproduction; **label lines** that do not cross keep a diagram readable; drawing the cross-section forces attention onto structure the child would otherwise never examine. By the end, the child knows the specimen better than anyone who merely ate one.

ACROSS K-5

K-1 draw whole specimens large and add color freely; skip the scale bar. Grades 2-3 add the cross-section and labels. Grades 4-5 take on the scale bar, the watercolor wash and the discipline of leaving highlights unpainted. Older children can also be asked to draw a specimen at two magnifications on the same sheet.

EXTEND IT Draw the same specimen once a week for three weeks as it ages, and bind the three drawings as a time series — the standard scientific-illustration move and a quietly powerful one. Grades 4-5: add a magnified detail inset with its own scale bar.

WATCH OUT An adult makes every cut. Pomegranate stains badly and permanently — cut it in a bowl of water, which also stops the juice spraying.

THE THREAD UNDERNEATH Looking properly is slow, and slowness is a skill that can be practiced.

The Plate Is a Frame

Composition — thirds, odd numbers, negative space, line

Same food, two plates. Why does one of them look like dinner and one look like art?

Grades 3–5 55 min Real mess No heat needed Needs the kitchen

YOU WILL NEED

- Components prepped in advance: sliced cucumber, cherry tomatoes, olives, herbs, cooked grains, a sauce in a squeeze bottle
- White plates of two sizes, plus one square or slate for comparison
- Tweezers, a small offset spatula, a pastry brush, a spoon for sauce swooshes
- A phone or tablet on a stand
- Printed reference sheets: three plates of the same food, composed three ways

WHAT YOU DO

1. Plate one: pile everything in the middle. Photograph it. This is the control.
2. Introduce three rules and try each on its own plate — put the focal point off center; use odd numbers; leave a third of the plate empty.
3. Build a plate that uses a line — a sauce swoosh, a row, an arc — to move the eye from one place to another.
4. Swap plates with another group and have them say, out loud, where their eye landed first and second.
5. Rebuild the best version on the bigger plate and then the smaller one. Note what changes.
6. Photograph all versions from the same height and lay them out side by side.

EVERYONE PARTICIPATES

The unit of work is arrangement, not eating, and the tweezers put a tool between the hand and the food for any child who wants that distance. Photographer, art director (decides the rules the plate will follow), plater and critic are four distinct jobs and only one of them touches food. The critic's job — saying where the eye goes — is the hardest and the most useful.

WORTH ASKING

- Where did your eye land first? Was that an accident or did someone put it there?
- What happened when we moved it to the smaller plate?
- Is there such a thing as too empty?

JOURNAL Plate-design page — sketch the plan before building it

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Plating is composition under three constraints a painter does not have: the picture is three-dimensional, it is made of things that wilt, and the viewer is about to destroy it. Those constraints make the standard compositional tools unusually legible.

Off-center placement creates tension where dead-center creates stillness; a plate is a frame and the rule of thirds works on it exactly as it works on a photograph. **Odd numbers** read as arranged and even numbers read as paired — three olives look composed, four look like a serving. **Negative space** is the one children resist hardest and the one that transforms a plate, because an empty rim reads as deliberate and a full plate reads as accidental. A **line** — a sauce stroke, a row of slices — gives the eye an entry point and a direction, and once children can say *where did your eye go first* about someone else's plate, they are doing genuine visual analysis.

Height and the plate's own size matter as much. The same arrangement that is elegant on a large plate is crowded on a small one, which is a lesson about the frame rather than the contents.

ACROSS K–5

Grades 4–5 run this as written. If K–1 are in the room, give them the same components and the single rule 'leave some plate showing', which they can do and which is the rule that matters most. Pair a younger plater with an older art director and require the director to use words only.

EXTEND IT Look at how food is composed in paintings and in restaurant photographs from several cultures and centuries, then make an original plate in a style the class invents and names. Do not copy a specific photograph or painting — the exercise is to identify the rules and then use them.

WATCH OUT Plated food that has been handled with tweezers and breathed on for forty minutes is a prop, not a meal. Decide before you start whether anything gets eaten, say so, and plan a separate clean portion if it does.

THE THREAD UNDERNEATH Care shows. Arranging something well is a way of saying the person receiving it matters.

Where Is the Light Coming From?

Photographing food — direction, diffusion, angle

Why does the same sandwich look delicious by the window and grim under the ceiling light?

Grades 3–5 55 min Light mess No heat needed Classroom is fine

YOU WILL NEED

- Tablets or phones, one per group of three
- One window with daylight, and the room's overhead lights for comparison
- White card and foil for reflectors, parchment paper or a thin white cloth for a diffuser
- Backdrops: a wooden board, a plain dish towel, a sheet of colored paper
- One prepared dish per group, and a small tripod or a stack of books

WHAT YOU DO

1. Photograph one dish under the ceiling light. Then the same dish beside the window. Compare on screen.
2. Move the dish so the window is behind it, then beside it, then in front of it. Three photographs, one variable.
3. Hold white card opposite the window to bounce light back into the shadows. Shoot with and without.
4. Tape parchment over the window and shoot again — hard light becomes soft light, and the difference is obvious.
5. Now change only the camera angle: overhead, three-quarter, and straight on at table level. Same dish, same light.
6. Pick the strongest frame from each group and lay them out together. Say what decision made each one work.

EVERYONE PARTICIPATES

The photographer never has to touch or taste the subject. This is one of the highest-status jobs available in the room and it requires nothing but eyes and decisions — which makes it a genuinely good standing role, as long as it rotates and does not become one child's permanent post.

WORTH ASKING

- Which photograph makes you look longest? Can you say what the light is doing in that one?
- We changed nothing but the angle. Why does it look like a different dish?
- Should a picture of food show how it really looked?

JOURNAL Print one photograph small and tape it in beside the sketch of the same dish

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Photography is a controlled-variable experiment that happens to produce pictures, and it is worth running it that way. Four variables do nearly all the work.

Direction. Light from behind the food rims it and makes anything translucent glow, which is why drinks and salads are usually backlit. Light from the side rakes across the surface and reveals texture — crumb, sugar crystals, the grain of bread. Light from the camera's position flattens everything, which is precisely why the overhead ceiling fixture and the phone flash both fail. **Diffusion.** A large soft source wraps around the subject and gives gentle shadow edges; a small hard source gives sharp black shadows. Parchment over a window makes the source larger and therefore softer. **Fill.** A white card opposite the light bounces some back and opens the shadows without adding a second light. **Angle.** Overhead suits flat, patterned food; three-quarter suits food with height; table level suits anything layered or stacked.

Worth naming honestly: professional food photography often uses props and substitutes that are not edible. This class photographs what it actually made, and that is a choice about truthfulness the children can understand.

ACROSS K–5

Grades 4–5 run the variables and hold the camera. K–2 are the light assistants — holding the reflector, moving the diffuser — and they can see the result change on screen as they move, which is the fastest possible feedback loop. Give the youngest children the job of choosing the backdrop; it is a real decision.

EXTEND IT Shoot the whole class cookbook. Grades 4–5: hold everything constant and vary only one thing across five frames, then lay them out as a labeled study — the standard way photographers actually learn.

WATCH OUT Agree the photography rules before any device comes out: only the food is photographed, faces only with permission, and nothing leaves the room without the school's usual media consent.

THE THREAD UNDERNEATH Every picture was made by someone who decided what you would see. Noticing that is media literacy.

Dough Is a Material

Sculpture — form, volume and the thing that holds its shape

Clay does what you tell it. Dough argues back. What is it trying to do?

Grades K-2 & 3-5 80 min plus baking Cover the table Adult at the heat Needs the kitchen

YOU WILL NEED

- A stiff bread dough — high flour, low water — enough for a fist-sized piece per child
- Salt dough, as a second material at the same table
- Scissors, forks, skewers, a small brush, and an egg wash or milk for glaze
- Baking sheets, parchment, and a hot oven
- Air-dry clay and a bowl form, for the vessel extension

WHAT YOU DO

1. Handle the dough before shaping anything. Push it, stretch it, let it spring back. Name what it does.
2. Make a coil, a ball, a slab, a pinch form. These are the four moves sculptors have always used.
3. Build a form that has to stand up. Decide where the weight goes before you build the top.
4. Cut, snip and score the surface for texture — scissors make spikes, a fork makes lines, a skewer makes holes.
5. Predict what will change in the oven. Write the prediction down before it goes in.
6. Bake, and compare the baked form with the prediction. Photograph both.

EVERYONE PARTICIPATES

Offer salt dough at the same table, in the same color, presented as a material choice rather than an alternative for anyone in particular — some children will prefer it because it is not food and others because it holds detail better. The sculpture is never required to be eaten; it is a made object, and treating it that way removes the whole question. Tools mean nobody has to touch dough with bare hands.

WORTH ASKING

- It went back to where it was. What is inside it that does that?
- What do you think the oven will do to this thin part? To this thick part?
- Where is the weight? What is holding it up?

JOURNAL Sketch the plan, then the prediction, then what actually came out

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Every material has a grammar, and learning it is what separates sculpting from squashing. Dough's grammar is gluten: an elastic protein network that resists being stretched and pulls back toward where it was. That is why a dough shape retracts and rounds itself off while you watch — and dough also sags, because unlike clay it has no stiffness of its own to hold a form against gravity. Clay stays where you put it; dough argues, and then sags. Rest the dough for ten minutes and it relaxes and becomes cooperative again — that is the gluten network releasing tension, and it is a genuinely useful thing for a child to feel in their hands.

Then the oven changes the material permanently. Steam inflates it, the gluten and starch set, the surface browns and hardens. Thin spikes crisp and darken, thick parts stay pale and soft, and undercuts close up as everything expands — which is why the prediction step matters. A sculptor working in dough has to think one transformation ahead, and that is a more sophisticated design problem than it looks.

The four moves — coil, ball, slab, pinch — are worth naming as the same four that ceramic traditions use worldwide, because the child is not doing a craft activity, they are doing what sculptors do with a material that happens to be in the kitchen.

ACROSS K-5

K-1 work the four basic moves and make solid forms. Grades 2-3 build forms that must stand. Grades 4-5 take on the prediction-and-comparison discipline and manage the trays in and out. Older children who finish early should be photographing, not making more.

EXTEND IT Vessels: make a pinch pot in air-dry clay and ask what a bowl has to do — hold, not leak, be liftable, be cleanable. Then use the finished bowls to serve in a later session, which closes a loop children genuinely feel.

WATCH OUT Raw flour should not be eaten and raw dough contains it — say this plainly at the start. Hot trays are adult-only in and out. Salt dough is not food at any point and gets a labeled tray of its own. Screen for wheat allergy and celiac disease before scheduling this card at all — airborne flour, not eating it, is the hazard, and a child with either cannot be in the room. Pour flour low, never sieve into the air, and wipe surfaces rather than sweeping. Salt dough is genuinely dangerous if eaten in quantity — the salt load is a real risk to a small child and lethal to a dog. Label the tray and tell families if any goes home.

THE THREAD UNDERNEATH Materials have opinions. Working with one means finding out what it wants to do.

Round and Round and Round

Radial symmetry, repeat units and pattern

Cut an orange across instead of down. What did you find?

Grades K-2 & 3-5 55 min Real mess No heat needed Classroom is fine

YOU WILL NEED

- Radially symmetrical produce: citrus, star fruit, okra, radish, kiwi, apple cut across the core
- Round plates or paper circles, and a compass or a circle template
- Thinly sliced components for building: cucumber rounds, carrot coins, pepper strips, herbs, seeds
- Tracing paper and pencils, for recording the repeat unit
- A mirror, for finding lines of symmetry

WHAT YOU DO

1. Cut produce across rather than lengthways. Line up the faces and count the segments in each.
2. Use the mirror to find the lines of symmetry. An orange has many; an apple cut across has five.
3. Draw the single repeat unit — the one wedge that the whole thing is made of.
4. Build a radial composition on a round plate: choose a repeat unit, decide how many times it goes around, work outward from the center.
5. Photograph from directly overhead. Off-axis ruins a radial design and the children will see it immediately.
6. Swap plates and count each other's repeats. Then find the mistakes — every radial pattern has one.

EVERYONE PARTICIPATES

Cutting, counting, arranging, photographing — no step requires tasting and the cut faces are the subject rather than the snack. A child who is uneasy about touching wet produce can work entirely in the drawing and counting role and complete the card fully.

WORTH ASKING

- How many times does it repeat? How do you know without counting every one?
- Where is the mirror line? Is there more than one?
- Where else have you seen a pattern like this that had nothing to do with food?

JOURNAL Pattern page, and the repeat-unit sketch

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Radial symmetry is a pattern the plant made for structural reasons and a pattern people have made deliberately for thousands of years, and putting the two side by side is the whole point of the card. Citrus segments are carpels; the apple's five-point star is its seed chambers; okra's star is the same. None of it was designed to be beautiful, and all of it is.

The vocabulary is genuinely mathematical. A **repeat unit** is the smallest piece from which the whole pattern can be generated; **rotational symmetry of order n** means the pattern maps onto itself n times in a full turn; a **line of symmetry** is where a mirror can stand. Children who can find the repeat unit in an orange can find it in floor tiles, in a rose window, in an Islamic geometric panel, in a Celtic knot, in a rangoli — radial pattern is close to a human universal and worth showing across cultures rather than from one tradition only.

The deliberate hunt for mistakes at the end is not a gotcha. Many pattern traditions include an intentional irregularity, and hand-made repeats always drift. Noticing drift is what makes a child look closely at a pattern rather than past it.

ACROSS K-5

This is one of the best genuinely mixed-age cards in the deck, because a kindergartener placing carrot coins around a circle and a fifth-grader working out rotational order are doing the same activity at two depths. Pair them deliberately: the older child names the order, the younger child builds it.

EXTEND IT Take the repeat unit into printmaking (card A03) and rotate one stamp around a center. Grades 4-5: build a pattern with translational symmetry — a strip that repeats sideways — and compare the two kinds of repeat.

WATCH OUT An adult cuts. Star fruit and citrus are slippery, and cutting across a round object is when a knife skids. Star fruit is a specimen here and is never eaten: carambola is dangerous for anyone with impaired kidney function.

THE THREAD UNDERNEATH People in many places, who never met, all found the same patterns beautiful. That is worth sitting with.

The Kitchen Orchestra

Sound, rhythm and a score anyone can read

Close your eyes in a working kitchen. How many different sounds can you name?

Grades K-2 & 3-5 50 min Light mess Adult at the heat Needs the kitchen

YOU WILL NEED

- The kitchen itself: whisk in a metal bowl, wooden spoon on a pot, rice in a jar, a box grater, a sizzling pan
- Dry goods in sealed jars — rice, lentils, popcorn, sugar — as shakers
- A tablet or recorder
- Large paper and markers, for the graphic score
- Optional: headphones, for children who want the room quieter

WHAT YOU DO

1. Two minutes of silence with eyes closed while one adult cooks. List every sound heard.
2. Sort the list: short or long, high or low, one hit or many, loud or soft.
3. Find or make an instrument for each category. A jar of rice and a jar of lentils are different instruments.
4. Invent a symbol for each sound. Not letters — shapes. Long sounds get long marks.
5. Write a score on the big paper: left to right is time, up and down is pitch. Four bars is plenty.
6. Perform it. Then swap scores with another group and perform theirs — the real test of whether the notation works.

EVERYONE PARTICIPATES

The most accessible card in the deck. There is no food contact of any kind unless a child wants it, and yet the subject is unmistakably food — so a child who is nowhere near tasting is doing the same real work as everyone else, with no accommodation visible to anyone. Offer headphones to the whole group as a standing option, not to anyone in particular.

WORTH ASKING

- How would you write down a sound that has no letters?
- Another group played your score. Did it come out the way you heard it in your head?
- Which foods are loud? Does loud tell you anything true about them?

JOURNAL Sound-words page, and the score itself folded in

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

This is a music and notation card and it earns its place in a food class for a specific reason: sound is the sensory channel with the lowest barrier to entry. A child can be an expert on the sound of a food without ever touching it, and crunch, sizzle, fizz and snap are genuine food properties that professional sensory panels measure.

The notation is the real content. Standard notation is a code that has to be taught; **graphic notation** is a code the children invent, and inventing one surfaces every problem real notation had to solve — how do you show two things at once, how long is long, how do you mark silence, how do you tell the reader how loud. A score that another group can perform is a score that works, and that swap is the assessment.

Composers have written for kitchen objects in earnest, and percussion traditions worldwide are built from tools and vessels rather than purpose-made instruments. Worth mentioning, because it means the children are not doing a novelty version of music.

ACROSS K-5

K-1 invent symbols and perform, which they do with more freedom than older children. Grades 2-3 organize the score left to right. Grades 4-5 handle simultaneous parts, dynamics and the conducting. Let a younger child conduct at least once — the authority is good for them and the older children cope.

EXTEND IT Record the performance and lay it under the stop-motion film from card T01. Grades 4-5: measure crunch properly — record five foods being bitten at the same distance and compare the waveforms on screen.

WATCH OUT Sizzling pans stay with an adult and the recorder goes to the pan, not the pan to the recorder. Set a firm ceiling on volume before you hand out anything that bangs, and mean it. Shakers are rigid plastic with taped lids, never glass — a failed lid puts lentils and popcorn kernels into K-1 hands. The box grater is not an instrument. For the eyes-closed listening, children sit at the far side of the room from the stove with an adult at the pan.

THE THREAD UNDERNEATH Somebody who cannot join in one way can lead in another. Build the room so that is always true.

What Shall We Call It?

Naming, typography and the identity of a dish

The same soup, on two menus, with two names. Which one would you order?

Grades 3–5 60 min Tidy No heat needed Classroom is fine

YOU WILL NEED

- Collected menus, food packaging, and market signs — a wide range of styles
- Good paper, black pens of several widths, colored pencils
- Tracing paper, rulers, pencils
- A tablet with a simple layout app, for grades 4–5
- The class's own dishes from earlier sessions, needing names

WHAT YOU DO

1. Spread out the collected menus. Sort them by how they feel before reading a word — fast, fancy, homely, loud.
2. Work out what is doing that: letterform, size, spacing, how much empty space, how many words.
3. Write one dish name six ways, changing only the lettering. Same words, different voice.
4. Now the words themselves. Name one class dish three ways: plain, descriptive, and with a story in it.
5. Build a one-page menu with a clear hierarchy — what should the eye read first, second, third?
6. Test it: hand it to a younger group and watch what they read first. Do not explain it. Just watch.

EVERYONE PARTICIPATES

No food at all. This is a drawing, writing and deciding session, which makes it a good one to schedule after a heavy cooking week. Every child names a dish, including one they have not tasted — describing a dish you have only smelled and looked at is a legitimate and quite difficult writing task.

WORTH ASKING

- You read that one first. Why? What made it win?
- Does this name tell someone what is actually in it?
- Same soup, two names. Is one of them dishonest?

JOURNAL Name-a-dish page, and the menu draft

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Typography is a set of decisions most people never notice and everyone responds to. Three ideas carry the whole session. **Hierarchy** is the order the eye takes things in, controlled by size, weight, position and space — a designer decides what is read first, and a page with no hierarchy is read in no particular order, which is to say not read. **Voice** is what a letterform implies before a word is decoded; a thin widely-spaced capital says expensive, a fat round one says cheerful, and the fact that the same three words can be made to say either is the most persuasive demonstration available. **Space** does more work than children expect, and margin is a design element rather than leftover.

The naming half is language arts in disguise and it is where the ethics live. 'Bean soup', 'white bean and rosemary soup' and 'the soup Ana's grandmother makes in October' are all true, and they set up completely different expectations. Naming honestly — telling the eater what is actually in it — matters especially in this class, where every ingredient is declared in advance. A name that hides an ingredient is the same failure as hiding the ingredient.

ACROSS K–5

Grades 4–5 handle hierarchy and layout. Grades 2–3 do the six-ways lettering exercise, which is where the concept lands. K–1 illustrate, and they should be the test readers at the end — watching a six-year-old read your menu is the most honest usability test a ten-year-old will ever get.

EXTEND IT Design the cover and contents page of the class cookbook (card T04) and print the covers from the vegetable blocks in A03. Grades 4–5: design a single dish label that must work at two sizes.

WATCH OUT Keep brand logos out of the making. Children can study real packaging and must design something of their own — copying a wordmark is not the exercise and it is not theirs to use.

THE THREAD UNDERNEATH Words and letters make promises. Keeping them is part of the design.

A Tool for This Hand

Designing for a real person, not an average one

How would you open that jar using only one hand?

Grades 3–5 80 min Light mess No heat needed Needs the kitchen

YOU WILL NEED

- Ordinary kitchen tools: peeler, jar, whisk, knife, can opener, measuring cup
- Constraint kit: a thick sock to wear on one hand, tape to bind two fingers, safety goggles smeared with petroleum jelly
- Build materials: card, foam, dowels, elastic bands, non-slip matting, zip ties, hot glue (adult)
- Non-slip mat, a board with a raised lip, an angled cup — as examples of real adaptive designs
- Sketch paper and a testing checklist

WHAT YOU DO

1. Try five ordinary kitchen jobs with one hand in a sock. Write down exactly where each one failed.
2. Do it again wearing the smeared goggles. Different failures, and now you have two lists.
3. Choose one failure. Write it as a problem statement: 'A person who ___ cannot ___ because ___.'
4. Sketch three solutions before building any of them. Argue about which to build.
5. Build a prototype. Test it under the same constraint that generated the problem.
6. Give it to someone else to test without explaining it. Watch what they do wrong. Fix that.

EVERYONE PARTICIPATES

No food is involved, and the card's subject is precisely that people's bodies differ and that the environment should adapt rather than the person. Running this early makes the rest of the course legible — when the class later serves components separately, or offers headphones to anyone, children recognize it as the same idea.

WORTH ASKING

- Where exactly did it fail? Not 'it was hard' — what physically could not happen?
- Your tester did it wrong. Is that their mistake or your design's?
- Who else would this help, besides the person you designed it for?

JOURNAL Problem statement, three sketches, and what the tester did

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Most kitchen tools are designed for a two-handed adult with steady grip and good vision, and that is a design decision rather than a fact of nature. The constraint kit lets children discover that quickly and in their own hands — a peeler needs one hand to hold and one to pull, a jar needs opposing force, a measuring jug needs you to read a line at eye level. None of these is inevitable, and every one has been solved by somebody.

The engineering content is **design for a stated user**, which is harder and more honest than design for yourself. It introduces the difference between a problem statement and a solution: 'jars are hard' is not workable, whereas 'a person with grip on one side only cannot hold the jar still while turning the lid' names the force that has to come from somewhere. Once that is on paper, non-slip matting, a wedge, a clamp and a lever all become obvious candidates.

The silent user test at the end is the professional move. Watching someone fail to use your prototype without being allowed to explain it is uncomfortable and is where almost all the learning is. It also teaches the idea underneath the whole card: a design that only works when the designer is standing next to it does not work.

ACROSS K–5

Grades 4–5 write problem statements and build. K–2 are the constraint testers and, crucially, the silent user testers — a seven-year-old handed a prototype with no explanation will find every flaw in ninety seconds. Make that their official job and give it status.

EXTEND IT Invite someone who uses adaptive equipment to visit and respond to the designs — arranged respectfully and in advance, as a consultation rather than a demonstration. Grades 4–5: redesign a tool so it works for both the constrained and unconstrained user with no switch between modes.

WATCH OUT Hot glue is adult-operated. Keep the constraint kit strictly a simulation of a task, never an impersonation of a person — the framing is 'this job is hard in this situation', never 'this is what it is like to be someone'. No blades and no heat under any constraint: the constraint trials use the jar, the whisk, the measuring cup, tongs and the can opener only, and anyone wearing the smeared goggles stays seated.

THE THREAD UNDERNEATH When a thing is hard to use, the thing is usually at fault. That is a surprisingly freeing idea.

Which Shape Holds the Sauce?

Pasta geometry, surface texture and a measurable test

Pasta comes in a hundred shapes. Is that decoration, or is it engineering?

Grades 3–5 70 min Real mess Adult at the heat Needs the kitchen

YOU WILL NEED

- Five or six pasta shapes: spaghetti, penne (ridged and smooth if possible), fusilli, farfalle, shells, orzo
- A thin sauce and a thick sauce
- A digital scale, a colander, identical bowls, a timer
- Pasta dough and a rolling pin, for the design half
- Graph paper

WHAT YOU DO

1. Cook equal weights of each shape. Drain them the same way for the same time.
2. Weigh 100 g of each cooked shape. Coat in the thin sauce, drain for exactly 30 seconds, weigh again.
3. Subtract. That number is how much sauce the shape held. Repeat with the thick sauce.
4. Graph the results — shape on one axis, sauce held on the other. Rank them.
5. Look at the winners. What do they have in common? Ridges, cups, twists, rough surfaces.
6. Now design your own shape in fresh dough, predict where it will rank, cook it, and test it the same way.

EVERYONE PARTICIPATES

The experiment is weighing and graphing; eating is separate and optional, and should be offered afterwards as an ordinary plate with components separate. Dry pasta shapes are also a good tactile material for anyone who wants to handle food that is unambiguously not about to be eaten.

WORTH ASKING

- What do the top three shapes have in common?
- Is there a best shape, or does it depend?
- What in our method might have made this unfair?

JOURNAL Bar graph of sauce held, plus the sketch of your own shape and its prediction

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Pasta shapes are a several-hundred-year-old engineering tradition and the design criteria are real: surface area, the presence of cavities that trap, surface roughness that sauce can key into, and how the shape behaves in a fork or a spoon. Ridged penne outperforms smooth penne for a reason a child can see and then measure — the ridges add surface and give the sauce something to grip. Twists and shells trap sauce inside themselves rather than merely wearing it.

The reason this card is worth running is that it converts a design question into a number. 'This one holds more sauce' is an opinion until somebody weighs it, and the weighing protocol has to be argued into fairness first: equal cooked weight, equal draining time, same sauce, same bowl. Children almost always want to skip the draining standardization, and the results are then meaningless, which is a good failure to have once.

Running it with both a thin and a thick sauce introduces the idea that there is no best shape, only a best shape for a purpose — thin sauces reward surface texture, thick ones reward cavities. That is a genuine engineering conclusion reached from the class's own data.

ACROSS K–5

K–1 sort the dry shapes, handle them, and describe them — an excellent vocabulary session in its own right. Grades 2–3 weigh. Grades 4–5 design the protocol, spot the unfairness, graph the results and shape the fresh dough.

EXTEND IT Test whether the same ranking holds for a very thin oil-based sauce. Grades 4–5: estimate surface area for two shapes and see whether it predicts the ranking — it partly will, which is a good conversation.

WATCH OUT Boiling pasta and draining are adult jobs; a colander of boiling water is one of the most common kitchen scalds. Cooked pasta that has been weighed and handled repeatedly is data — cook a separate clean batch if you intend to eat.

THE THREAD UNDERNEATH Someone designed this, on purpose, a long time ago. Most ordinary things are like that.

The Lunchbox Problem

Structure, moisture and a sandwich that survives until noon

It was fine at eight in the morning. Why is it a disaster at twelve?

Grades K-2 & 3-5 60 min plus a four-hour wait Real mess No heat needed Needs the kitchen

YOU WILL NEED

- Bread of two or three kinds, plus wraps and crackers
- Wet fillings — tomato, cucumber, sauce — and barrier candidates: hummus, seed spread, oil-based spread, lettuce
- Containers of several kinds, plastic wrap, foil, paper
- A backpack, and a defined shake test
- A scale, and a camera

WHAT YOU DO

1. Name the failure modes first, from experience: soggy, squashed, leaked, fell apart, went warm.
2. Pick one to solve. Write the criterion so it can be judged: 'the bread under the tomato is still dry at noon'.
3. Build four versions of the same sandwich, changing one thing each time — barrier layer, bread, order of layers, container.
4. Weigh each one and pack them all. Same backpack, same twenty shakes, same shelf.
5. Four hours later, open them in front of everyone. Photograph the cross-sections.
6. Score against the criterion and identify which change did the work.

EVERYONE PARTICIPATES

The work is building, packing, waiting and assessing. Children build the sandwich they would actually want, which means the components are chosen by them and nobody is handed an unfamiliar filling. Whether anyone eats the result is beside the point — after four hours in a backpack these are specimens.

WORTH ASKING

- Which failure are we solving? You cannot solve all four at once.
- What is the barrier actually doing?
- Would your solution survive a bus ride as well as a backpack?

JOURNAL Cross-section drawings at 0 hours and 4 hours

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Sogginess is moisture migration, and it obeys the same rule as everything else that moves on its own: water travels from wet to dry until things even out. A tomato slice against bread is a wet thing against a dry porous thing, and bread's structure is full of spaces that pull water in. A **barrier layer** works by putting something water cannot easily cross in between — fat is excellent, which is why butter or cream cheese spread edge to edge outperforms a thicker bread; a lettuce leaf is a physical shield; a cheese slice is both.

Layer order matters for the same reason: wet ingredients in the middle, dry and fatty ones against the bread. **Structure** is a separate failure mode entirely — a sandwich collapses because the layers slide, and friction, a smaller footprint, a tighter wrap or a rigid container all fix that. Naming the failure modes separately is what keeps children from changing five things at once.

It is worth saying what makes this card land: every child in the room has direct personal experience of the failure, most of them have a strong opinion about it, and the solution is one they can use tomorrow.

ACROSS K-5

K-2 name the failure modes, which they do vividly, and run the shake test. Grades 2-3 build the versions. Grades 4-5 hold the one-variable discipline and run the scoring, and they should be the ones who notice when a younger partner has changed two things at once.

EXTEND IT Run it on wraps and crackers and see whether the ranking holds. Grades 4-5: weigh each sandwich before and after to measure how much water moved, and connect it to osmosis on card S01.

WATCH OUT Anything with meat, egg, dairy or mayonnaise that sits out for four hours is a food-safety problem, not an experiment — use vegetable and spread fillings only, keep the test at room temperature and short, and throw everything away afterwards. Say plainly at the start that these will not be eaten. The school supplies every bread, filling, container and bag. Never use a child's own lunch or lunchbox as material or example and never compare real lunches — the failure modes are a general design problem and stay general.

THE THREAD UNDERNEATH A problem you have complained about for years turns out to be solvable in an afternoon.

What Kneading Actually Does

Dough as a material, and a test you can hold up to the light

Two bowls, same ingredients. One is shaggy and one is silky. What happened in between?

Grades 3–5 75 min Cover the table Adult at the heat Needs the kitchen

YOU WILL NEED

- Bread flour and cake or pastry flour, water, salt, yeast
- A scale, bowls, a bench scraper, a timer
- A bowl of water and a sieve, for the gluten-washing demonstration
- Loaf pans or a baking sheet
- A ruler, for measuring rise

WHAT YOU DO

1. Mix two identical doughs. Knead one for ten minutes. Leave the other mixed but unkneaded.
2. Every two minutes, stretch a piece of the kneaded dough thin and hold it to the light. Record when it goes translucent without tearing.
3. Poke both doughs. Stretch both. Describe the difference using the texture word wall.
4. Mix a third dough with cake flour and knead it the same ten minutes. It will never get there. Ask why.
5. Wash a lump of kneaded dough gently under water in a sieve until the water runs clear. What is left is gluten.
6. Prove and bake all three. Measure the rise and cut them open to compare the crumb.

EVERYONE PARTICIPATES

Dough handling is optional and there are plenty of jobs for anyone who would rather not — the timer, the windowpane photographer, the measurer of rise, the recorder. Offer disposable gloves openly to the whole group rather than to anyone in particular; some children want them for mess and some for texture and nobody needs to say which.

WORTH ASKING

- What changed between minute two and minute ten? Nothing was added.
- Why can this flour not do it, however hard we work?
- What is left in the sieve, and what left with the water?

JOURNAL Windowpane timeline, plus a drawing of the crumb of each loaf

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Flour contains two proteins that do nothing much on their own. Add water and they begin to link; work the dough and they organize into a continuous elastic network — gluten — that can stretch thin without breaking and can trap gas. The **windowpane test** is a direct read on that network: stretch a piece thin enough to see light through it, and if it holds without tearing the network is developed. Children can watch it arrive, two minutes at a time, which makes an invisible process into something with a timestamp.

The cake-flour dough is the control that proves it is the protein and not the effort. Cake and pastry flours are milled from softer wheat with less of these proteins, so no amount of kneading produces the same network — which is exactly why they make tender cakes and terrible bread. The washing demonstration is the clincher: rinse the starch away and a springy gray lump of nearly pure gluten is left in the sieve, which children can stretch like elastic.

The engineering frame is material science. A material's properties come from its internal structure, structure can be changed by processing, and the right material for a job depends on what the job demands. That sentence is true of steel and of dough.

ACROSS K–5

Kneading is physical and fun and K–2 do it enthusiastically in short bursts — run a relay so the dough keeps moving. Grades 4–5 run the windowpane timing, the flour comparison and the gluten washing, and measure the crumb at the end.

EXTEND IT Compare kneaded bread with a long slow no-knead dough and discover that time can substitute for work. Grades 4–5: weigh the washed gluten from bread flour and from cake flour and report it as a percentage of the starting dough.

WATCH OUT Raw flour and raw dough are not to be tasted, and raw flour is a real food-safety risk rather than a technicality. Ovens are adult-operated. Wash the gluten over a bowl, not the sink — flour paste blocks drains. Screen for wheat allergy and celiac disease before scheduling this card at all — airborne flour, not eating it, is the hazard, and a child with either cannot be in the room. Pour flour low, never sieve into the air, and wipe surfaces rather than sweeping.

THE THREAD UNDERNEATH Ten minutes of steady work changes a material completely. Not everything gives feedback that clearly.

Making Water Clear Is Not Making It Safe

Filtration, and the difference between clean and clear

Your filter made the muddy water look like tap water. Would you drink it?

Grades 3–5 70 min Cover the table No heat needed Needs the kitchen

YOU WILL NEED

- Plastic bottles with the bottoms cut off, to invert as filter columns
- Gravel, coarse sand, fine sand, activated charcoal, cotton wool, coffee filters
- Muddy water made with soil, a little cooking oil, and food coloring
- Clear jars, a measuring cylinder, a timer
- A torch, to judge clarity against

WHAT YOU DO

1. Make one batch of dirty water and use the same batch for every group, so results can be compared.
2. Each group designs a layered column. Sketch the layers and predict, in order, what each will catch.
3. Pour a measured volume through. Time how long it takes and measure how much comes out.
4. Judge the result three ways: clarity against a torch, color, and smell.
5. Redesign once. Most groups will reorder their layers, and they should say why.
6. Then the real question: the water is clear. Is it safe? Work out what a filter like this cannot remove.

EVERYONE PARTICIPATES

No food and no tasting of any kind, by rule. The card is building, pouring, timing and judging, and it suits children who want to be hands-on with something that could not possibly be offered to them to eat.

WORTH ASKING

- Which layer caught what? How can you tell?
- Your filter is slow and your neighbor's is fast. Which is better?
- It looks clean. What might still be in there?

JOURNAL Layer diagram with predictions, plus a before-and-after clarity note

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

A layered filter works by sieving at several scales at once. Gravel catches the big debris, coarse sand catches smaller particles, fine sand catches smaller ones again, and cotton or paper catches the finest. Order matters enormously — coarse to fine, top to bottom — and a group that puts fine sand on top will find their filter clogs and stops, which is a better lesson than being told. **Activated charcoal** works differently: rather than sieving, it has an enormous internal surface to which dissolved molecules stick, which is why it takes out color and smell that sand cannot touch.

And then the honest ending, which is the whole reason this card belongs in a food course. **Clear is not safe.** Bacteria, viruses and dissolved chemicals pass straight through a sand-and-charcoal column, and the water that comes out may look perfect and still make someone very ill. Making that the punchline rather than a footnote is the point: children learn that their senses are not a safety test, which is the same principle behind every food-safety rule in the kitchen.

Real drinking-water treatment adds steps this model cannot — flocculation, then disinfection by chlorine, ultraviolet light or boiling — and that is worth showing as a diagram so children see what their filter is missing.

ACROSS K–5

K–2 build columns and pour, and they are excellent at judging clarity. Grades 4–5 design the layer order, measure flow rate and volume recovered, and lead the clear-versus-safe discussion — which they should explain to the younger children in their own words.

EXTEND IT Compare flow rate against clarity and find the trade-off — the best-filtering column is usually the slowest. Grades 4–5: research how the school's own drinking water is treated and diagram the real process next to their model.

WATCH OUT Say it at the start and again at the end: **nobody drinks anything from this activity, ever**, no matter how clear it looks. Use soil rather than anything organic or rotting, wash hands thoroughly afterwards, and keep this activity's equipment separate from food equipment. Use bagged sterile topsoil or play sand, never garden or playground soil. Activated charcoal is aquarium grade, rinsed first, never poured dry from a height — the dust is a respiratory irritant. An adult cuts every bottle and tapes the cut edge.

THE THREAD UNDERNEATH Looking fine and being fine are different. Knowing the difference is most of safety.

Design This Kitchen

Workflow, stations and how far a person has to walk

Why does everything take so long when six people are cooking?

Grades 3–5 65 min Light mess No heat needed Needs the kitchen

YOU WILL NEED

- A simple recipe the class has already made once
- Graph paper and a scale drawing of the actual kitchen, prepared in advance
- String or a trundle wheel, to measure distance walked
- Sticky notes, one per task
- A stopwatch and a tally counter

WHAT YOU DO

1. Run the recipe once the normal way. One person's only job is to follow a cook with the string and measure the distance they walk.
2. Count the collisions — every time two people have to wait for each other or for a piece of equipment.
3. Write every task on a sticky note. Arrange them on the floor plan where they actually happened.
4. Redesign: move stations, split a bottleneck, put the tools where the task is. Argue it out.
5. Run it again with the new layout and measure the same two numbers.
6. Report: distance walked before and after, collisions before and after. Did it help?

EVERYONE PARTICIPATES

Measuring, counting, mapping and redesigning — no food contact is required for any of it. The person with the string and the person with the tally counter have the two most important jobs in the session.

WORTH ASKING

- Where did people keep waiting? Was it a person or a thing?
- How far did you walk? Was any of it necessary?
- Does the new layout work for someone who cannot reach the top shelf?

JOURNAL Floor plan before and after, with the two numbers

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Professional kitchens are designed, not arranged, and the design principles are simple enough for children to derive from their own data. The classic idea is that the three things a cook returns to constantly — cold storage, water, and heat — should form a compact triangle, because every step between them is repeated dozens of times a session. Put the trash can far from the prep board and you have built a hundred extra meters of walking into every class.

Two measurements make the argument. **Distance walked** is easy to collect and startling: children routinely find one person walked several hundred meters in a forty-minute session. **Collisions** — the number of times someone had to wait for a person or a tool — locates the bottleneck, and a bottleneck is almost always a shared resource rather than a slow person, which is a point worth making explicitly and kindly.

The redesign is genuine systems engineering: tools stored at the point of use, tasks that can run in parallel separated in space, sequential tasks placed adjacent, and one path in and a different path out so that traffic does not meet itself. Every one of those is a real principle, and a class that has derived them owns its own room in a way it did not before.

ACROSS K-5

K-2 do the walking measurement, which they love, and place sticky notes. Grades 4–5 draw the plan to scale, run the analysis and chair the redesign argument. Insist that the redesign has to work for the shortest person in the room as well as the tallest — reach is a real constraint and the younger children should raise it.

EXTEND IT Adopt the winning layout for the rest of the term and post the map. Grades 4–5: time the same recipe in both layouts and report the difference as a percentage.

WATCH OUT Any layout change has to be checked against the school's actual safety requirements — exits, fire equipment, hot surfaces and traffic past them are not the children's to move. Frame that as a design constraint rather than a veto.

THE THREAD UNDERNEATH When something is slow and frustrating, it is often the room's fault rather than anyone's.

Twelve Pictures a Second

Stop-motion animation of a recipe

How do you show a whole afternoon of cooking in thirty seconds?

Grades K-2 & 3-5 90 min over two sessions Real mess No heat needed Needs the kitchen

YOU WILL NEED

- Tablets with a stop-motion app, one per group of three
- A stand or a stack of books to hold the camera absolutely still
- A lamp, so the light does not change when a cloud goes over
- Ingredients that can be moved incrementally: chopped vegetables, flour, seeds, dough
- Storyboard sheets — six boxes per group

WHAT YOU DO

1. Storyboard first: six boxes, six key moments. No filming until the boxes are drawn.
2. Fix the camera. Tape the stand down. If the camera moves, the film is ruined and the group will discover this.
3. Shoot a test: move one object a tiny amount, take a frame, repeat ten times. Play it back.
4. Work out the rule for yourselves — small moves and many frames give smooth motion; big moves give a jumpy one.
5. Shoot the real sequence. Someone has to count frames aloud and someone has to keep hands out of shot.
6. Add the sound from the kitchen orchestra (card A09) underneath, and a title card.

EVERYONE PARTICIPATES

Nothing is tasted and the food is a puppet. Jobs: storyboarder, camera operator, frame counter, animator, continuity watcher and sound designer — six roles, none of which requires eating and only one of which touches food. A child who moves each pea a few millimeters for an hour is doing the most demanding job in the room.

WORTH ASKING

- Your film is jumpy. What would fix it, and what would that cost you?
- Something flickers. What changed between frames?
- What is the one moment this film has to show?

JOURNAL Storyboard page — draw the six boxes before filming

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Animation works because of how vision and attention handle a fast series of still images: shown quickly enough, the brain reads them as continuous motion rather than as separate pictures. That is the whole technology, and it was the whole technology in 1900 as well.

The craft rules all follow from one number — frames per second. At twelve frames per second, one second of film needs twelve photographs, so a thirty-second film needs three hundred and sixty of them. Children invariably underestimate this and then have to solve it, usually by making the movements bigger, which makes the motion jumpier. That trade-off between labor and smoothness is the real lesson and no amount of explaining substitutes for hitting it.

Two disciplines matter more than anything else: the camera must not move, and the light must not change. Both sound trivial and both are where amateur stop-motion falls apart, because a shifting shadow across three hundred frames flickers horribly. Taping the stand down and using a lamp rather than a window is what separates a film from a mess, and it is a lesson about controlling variables in a completely different context.

ACROSS K-5

K-1 are the animators — small repetitive precise movements are well within them and they are patient about it. Grades 2-3 storyboard. Grades 4-5 run the camera, count frames, and edit. Pair deliberately, because the counting and the moving have to stay in sync.

EXTEND IT Screen the films at family night. Grades 4-5: animate a process that is genuinely invisible in real time — dough rising over two hours, shot one frame a minute.

WATCH OUT Agree the filming rules first: food and hands only, no faces without the school's usual consent, nothing leaves the building without permission. Food handled for ninety minutes under a lamp is a prop and is composted afterwards.

THE THREAD UNDERNEATH Three hundred small identical efforts add up to something nobody could have done in one go.

Ask, Then Stop Talking

Recording an interview about a food that matters to someone

Who taught the person who taught you?

Grades 3–5 60 min plus interviews at home Tidy No heat needed Classroom is fine

YOU WILL NEED

- Recorders — tablets or phones — one per pair
- Headphones, for listening back
- Question-planning sheets
- A consent slip for the person being interviewed, in the languages your families use
- A quiet corner, or a coat cupboard, which is the best recording booth a school has

WHAT YOU DO

1. Listen to a short well-made interview. Notice how little the interviewer says.
2. Sort question types: ones answered yes or no, and ones that cannot be. Cross out all the first kind.
3. Write six open questions about a dish someone in your family or community makes.
4. Practice on a partner. Record it. Listen back and count how many times you interrupted.
5. Learn the hardest technique: after they finish, wait three full seconds before speaking. Practice it.
6. Interview for real, with consent recorded at the start. Bring the audio back and choose the ninety best seconds.

EVERYONE PARTICIPATES

There is no food in this session at all. It is also the card that most reliably brings a quiet child forward, because the assignment is to listen rather than to perform, and because the expert being interviewed is somebody who loves them.

WORTH ASKING

- Which of your questions can be answered with one word? Fix those.
- What did they say after the pause?
- Whose story is this, and who gets to decide what happens to the recording?

JOURNAL Interview page — the questions, and one sentence you want to keep

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Interviewing is a technical skill with a small number of rules, and children can get good at it quickly because the rules are concrete. **Open questions** — 'what do you remember about' rather than 'did you like' — are the whole game; a yes-or-no question gets a yes or a no and then silence. **The three-second wait** is the technique professional interviewers rely on most, because people almost always add something better after the pause, and children find it excruciating and then see it work. **Not interrupting** is measurable: record a practice run and count.

The technical side is real too. Recording close to the speaker in a soft-furnished room beats a beautiful microphone in a hard one, because most bad audio is reflections rather than equipment. Test levels first. Record thirty seconds of room tone with nobody speaking, which every editor needs and nobody remembers.

And consent is part of the technology, not an add-on: you ask before recording, you record the person saying yes, you say who will hear it, and you agree to delete it if they later want that. Children who learn this at nine have a framework for a world that will ask them to record things constantly.

ACROSS K–5

Grades 4–5 write and conduct. K–2 can interview with two questions and an adult holding the recorder, and they ask better questions than older children because they have not learned to be polite about it. Pair a younger child as the second interviewer with one question of their own.

EXTEND IT Cut the best moments into a class audio piece for family night, with permission from every speaker. Pair with card C01, where the dish itself gets cooked.

WATCH OUT Consent is not optional and not implied. Use the school's process, offer the slip in the family's own language, and make it genuinely easy for a family to decline — some will have good reasons that are none of the school's business. Never require a child to produce a family story; offer a school staff member or a neighbor as an equally good subject. Recorders are lent by the school, and every interview can be done at school, in school hours, with a staff member or a visitor as the subject. Written notes are an equal alternative to audio. Nobody's next session depends on what came back in a bag.

THE THREAD UNDERNEATH Listening properly is a thing you can get better at, and most people never try.

The Cold Chain

Thermometers, data logging, and the gap food must not sit in

Your lunchbox left home cold. Was it still cold at lunchtime?

Grades 3–5 55 min plus a logging day Light mess No heat needed Needs the kitchen

YOU WILL NEED

- Several instant-read or probe thermometers, ideally one per group
- Insulated bags, ice packs of different sizes, an ordinary lunch bag as the control
- Water bottles filled to identical weights, to stand in for food
- A recording sheet ruled for time and temperature, and graph paper
- A fridge and freezer thermometer, to check the school's own

WHAT YOU DO

1. Calibrate first: every thermometer into stirred ice water, which is 0°C. Write each one's error on a tape flag.
2. Set up four packed bags that differ in exactly one way each, all starting at the same temperature.
3. Log the temperature of each every thirty minutes across the school day. Assign the job and keep it.
4. Plot all four as lines on one graph, time along the bottom.
5. Shade the band between about 5°C and 60°C. This is the range where food-safety guidance says not to leave food sitting.
6. Read off the time each bag entered the shaded band. That is the answer to the question.

EVERYONE PARTICIPATES

Water bottles stand in for food, deliberately, so the experiment involves nothing edible. Calibrating, logging, plotting and reading the graph are the work, and the logger who remembers every half hour is the most valuable person in the class that day.

WORTH ASKING

- Two thermometers, same water, different numbers. Which do we believe?
- Whose bag stayed out of the band longest? What did they do differently?
- Why does a bigger ice pack help, when it is the same temperature as a small one?

JOURNAL Line graph page with four lines and the shaded band

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

A lunchbox is a thermal system and a data-logging problem at the same time. Heat always moves from warmer to cooler, so nothing keeps cold in — insulation slows heat coming in, and an ice pack absorbs a large amount of heat while melting without itself getting warmer, which is why a bigger pack buys more time rather than a lower temperature.

The graph is where the learning is, and this is a card about line graphs rather than bar charts: time on the horizontal axis, temperature on the vertical, one line per condition. Real data is messy, the lines cross, and somebody will forget a reading — all of which is worth leaving in and discussing rather than tidying up.

The shaded band turns the graph into a decision. Food-safety guidance treats roughly 5°C to 60°C as the range in which bacteria multiply fastest, and the practical rule is to limit how long food sits there. Children who have watched their own test bag cross that line at 10:40 in the morning understand refrigeration, ice packs and the whole global cold chain of trucks and containers differently. Calibrating the thermometers first is not housekeeping: an uncalibrated instrument would make the entire graph a guess.

ACROSS K–5

K–2 read thermometers and call out numbers, which is real practice at reading a scale. Grades 4–5 calibrate, plot and interpret. Pair them for the logging rounds so the reading and the recording are done by two different people, which is also how scientists do it.

EXTEND IT Log the school fridge for a week. Grades 4–5: work out how much ice is needed for a six-hour trip and test the prediction.

WATCH OUT Use water bottles, not real food, for anything that will sit out — this is a temperature experiment, not a spoilage one, and deliberately warming actual food to study bacteria is not a school activity. Probe thermometers are sharp and are washed between uses. The school supplies the bags; never use a child's own lunch or lunchbox. Teach the band as physics, not as a verdict — never tell a child, or let a child conclude, that their own lunch is unsafe.

THE THREAD UNDERNEATH An invisible thing was happening all morning, and a number made it visible.

Make the Book

Layout, editing and publishing the class cookbook

How does a pile of paper become something a stranger can follow?

Grades 3–5 Three sessions across the term Tidy No heat needed Classroom is fine

YOU WILL NEED

- Tablets or computers with any layout or document tool
- Every recipe the class has made, in whatever state it is in
- The photographs from card A06 and the prints from card A03
- A printer, and something to bind with
- A QR-code generator, if the school hosts the audio from card T02

WHAT YOU DO

1. Agree the template first. One page per recipe, the same shape every time: title, ingredients, steps, a note.
2. Editing pass: does every recipe list every ingredient? Numbers and units on everything?
3. Test each recipe on a reader who has never made it — grades 4–5 hand theirs to K–1 and watch them try to follow it.
4. Fix what the test found. This is almost always missing quantities and missing 'meanwhile'.
5. Lay out the pages with the photographs and the printed endpapers. Add page numbers and a contents page.
6. Print, bind, and put a copy in the school library as well as one in every child's bag.

EVERYONE PARTICIPATES

No food. This is writing, design, testing and assembly, and it produces the one object every child takes home with their own name in it. A child who cooked little and drew a great deal is as fully represented in the book as anyone.

WORTH ASKING

- Your tester stopped here. What is missing at that exact point?
- Does the list say everything the method uses?
- Who is this book for, and what do they already know?

JOURNAL My recipe page — write one from memory before editing it

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

A recipe is technical writing for an audience that cannot ask questions, which is a genuinely difficult specification and a useful one for children to meet. Three things make it work.

Consistency of structure. Every recipe in the same order, with the same headings, so a reader learns the shape once and then navigates any page. That is why every cookbook ever printed looks broadly alike, and children who design their own template understand it as a decision rather than a convention.

Completeness. The commonest failure is an ingredient that appears in the method but not in the list, and the second commonest is a quantity given as 'some'. A user test catches both instantly. Handing a recipe to a six-year-old and watching without speaking is the most efficient editing tool available.

Honesty about ingredients. In this class every ingredient is named — that is the same rule that governs the kitchen itself, and it carries into the book. A recipe that hides something is a recipe that will hurt somebody eventually, and children can hold that standard.

ACROSS K–5

Grades 4–5 write, edit and lay out. K–1 are the user testers, which is the highest-leverage role in the whole process, and the illustrators. Grades 2–3 proofread quantities against methods, which is a checkable job with a definite right answer.

EXTEND IT Add QR codes linking to the stop-motion films and the interview audio, if the school hosts them. Grades 4–5: produce a large-print version, and discuss who that helps.

WATCH OUT Photographs of children need the school's usual consent. Family recipes are shared with the family's permission and credited by name if they want that, and by no name if they do not.

THE THREAD UNDERNEATH Writing something down so a stranger can do it is harder and more generous than doing it yourself.

Before Anyone Owned a Thermometer

How people cooked precisely without precise instruments

A recipe from 1850 says 'a moderate oven'. How did anyone know?

Grades 3–5 55 min Light mess Adult at the heat Needs the kitchen

YOU WILL NEED

- Historical recipes — genuinely old ones, with vague quantities and no temperatures
- Flour, for the oven-test method, and a modern oven thermometer for comparison
- A song everyone knows, for timing
- String, a handspan, a teacup, a spoon — the old measures
- The same modern recipe written two ways: by weight, and in old-fashioned terms

WHAT YOU DO

1. Read a historical recipe aloud. List everything it does not tell you.
2. Time something by singing rather than by clock. Three verses. Compare across groups — how close were you?
3. Measure the same flour by teacup, by handful, by spoon and by scale. Compare the spread.
4. Test the old oven method: spread a spoon of flour on a tray, leave it exactly five minutes, and read the heat from the color — light, golden or dark. Check it against a thermometer.
5. Cook one thing entirely by the old methods and one by modern measurement. Compare the results.
6. Discuss: what did precision cost, and what did it buy?

EVERYONE PARTICIPATES

Measuring, singing, timing and comparing. The song-timing round is a great equaliser and involves no food at all. Whether anything gets eaten at the end is a separate and optional matter.

WORTH ASKING

- How much did the handfuls differ? Whose hand is the standard?
- What can a person tell about an oven without a thermometer?
- What did we gain when recipes got exact? What did we lose?

JOURNAL Measurement-spread page — record every handful

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Standardized measurement in home cooking is surprisingly recent. Recipes for most of history assumed a cook who had been taught by another cook and could judge heat, thickness and doneness by sight, sound and feel. Quantities were relational — a teacup of this to two of that — and heat was named by what it did: a spoon of flour spread on a tray for five minutes came out light brown in a slow oven, golden in a moderate one and dark brown in a hot one. The test still works, and it is worth running against a thermometer — not because it is precise, but because three broad bands turn out to be enough to bake by.

The shift to level, standardized measures in the late nineteenth century, most famously promoted by Fannie Farmer's 1896 cookbook, changed what a recipe was for: it made a recipe transmissible to a stranger who had never watched anyone cook. That is a technology story about writing and instruments, not about cooking, and it is the same story as the class cookbook in card T04.

The trade-off is worth drawing out honestly. Standardization buys reliability and reach; it costs the judgement that came from having to read the food rather than the clock. A cook who only follows numbers is helpless when the numbers are wrong, and the flour-in-the-oven test is a reminder that senses are instruments too.

ACROSS K–5

K–2 do the handspan and teacup measuring and the singing, and they are better at judging 'is it done' by looking than they are given credit for. Grades 4–5 run the flour oven test against the thermometer and compute the spread between the non-standard measures.

EXTEND IT Connect to the kilogram itself, which is no longer a metal cylinder in a vault but is defined by a constant of nature — a good story about why anyone bothers. Grades 4–5: rewrite a modern recipe in 1850 language and have another group try to cook from it.

WATCH OUT The oven flour test is adult-operated, with a timer and a tray at a known rack position, and the oven door is opened by an adult only.

THE THREAD UNDERNEATH People were extremely skilled before they had instruments. Different, not worse.

Count the Spirals

Fibonacci numbers, growing things, and a pattern nobody planned

Count the spirals going left. Now count the ones going right. Why those two numbers?

Grades 3–5 55 min Light mess No heat needed Classroom is fine

YOU WILL NEED

- Pineapple, romanesco, artichoke, pinecones, a sunflower head if the season allows
- Colored stickers or washable markers, for tracing spirals without losing count
- Squared paper, and a compass
- A calculator for grades 4–5
- A printed Fibonacci sequence, kept hidden until the counting is done

WHAT YOU DO

1. Pick a specimen. Find one spiral going clockwise and mark every scale along it with a sticker.
2. Count how many parallel spirals there are in that direction. Then do the anticlockwise ones.
3. Write both numbers on the board with the specimen's name. Collect every group's pair.
4. Now reveal the sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, 55. Look at the board.
5. Find the rule that generates the sequence. Then check it against every number on the board.
6. Grades 4–5: divide each number by the one before it, all the way along, and watch what the answers do.

EVERYONE PARTICIPATES

Counting, marking and recording — the produce is a mathematical object here and nobody is going to be offered a piece of pineapple. A child who finds food overwhelming can hold and count a pineapple or an artichoke for an hour quite happily.

WORTH ASKING

- What is the rule that makes the next number?
- This one gave 7 and 11, which are not in the sequence. What should we do with that?
- Did the plant choose these numbers?

JOURNAL Spiral-counting page — draw the specimen and label both counts

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Count the spirals on a pineapple, a pinecone or a sunflower head and the two counts usually turn out to be neighboring numbers in the sequence where each term is the sum of the two before it — 8 and 13, or 13 and 21, or 34 and 55. This is one of the few places where a striking mathematical pattern falls out of an ordinary object on a table, and the moment the board is compared with the hidden sequence is genuinely memorable.

It happens because of how these plants grow. New parts are added one at a time around a growing tip, each one turned by a consistent angle from the last. When that angle is close to a particular irrational proportion, successive parts never line up and so pack more efficiently — and the visible consequence of that packing is spiral families whose counts land on that sequence. The plant is not doing arithmetic; the arithmetic is a consequence of the packing.

Two honesties are worth keeping. Not every specimen conforms — real pinecones sometimes give counts from a related sequence, and damaged or hybridised specimens can give nothing recognizable at all. Collect the exceptions rather than discarding them. And the ratio exercise for older children is lovely precisely because the answers settle down without ever repeating: divide each term by the one before and the results converge on about 1.618, approaching but never landing.

ACROSS K–5

K–1 trace spirals with stickers, which is the hardest physical part and which they enjoy. Grades 2–3 count and record. Grades 4–5 find the generating rule and run the ratio calculation. Pair across — the counting is much easier with two people.

EXTEND IT Build a Fibonacci spiral on squared paper from squares of 1, 1, 2, 3, 5, 8 and lay a real specimen beside it. Grades 4–5: chase the ratio to ten decimal places and see it stop changing much.

WATCH OUT Pinecones and artichokes are prickly. Bracts on a globe artichoke have sharp tips — snip them first or use pinecones for the youngest children.

THE THREAD UNDERNEATH A pattern was sitting in the fruit bowl the whole time, and now you can see it.

Cooking for Two Hundred

Scaling up, division with remainders, and what a tray holds

This makes twelve. The whole school is three hundred and forty. Now what?

Grades 3–5 60 min Tidy No heat needed Classroom is fine

YOU WILL NEED

- A recipe the class has made, written for a small number
- The real roll number for the year group or school
- Actual tray and pan dimensions from the school kitchen, and the oven's capacity
- Calculators, and supplier prices in bulk sizes
- Large paper for working

WHAT YOU DO

1. Get the real number of people from the office. Use the real number, not a round one.
2. Work out the scale factor. It will not be a whole number, and that is the point.
3. Scale every ingredient. Convert anything that has become silly — 47 teaspoons is a teaspoon short of a cup.
4. Now the packaging problem: flour comes in fixed bag sizes. How many bags? What is left over?
5. Now the oven problem: how many trays, how many batches, how long in total? Does it fit in the lunch hour?
6. Present the plan — quantities, shopping list, timings, and what you would do differently.

EVERYONE PARTICIPATES

This is a paper-and-calculator session with no food present. It is also a good one to schedule when the kitchen is unavailable, and it gives children who are strong at arithmetic a chance to lead in a course that has mostly rewarded other things.

WORTH ASKING

- Why do we round the flour up and the servings down?
- What does 7.4 trays actually mean?
- Does your plan fit in the time the kitchen has?

JOURNAL Scaling page — the working, not just the answer

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Scaling a recipe for a real institution is where arithmetic stops being tidy, and that is exactly why it is worth doing. Three kinds of division show up and they behave differently.

Proportional scaling is straightforward multiplication, but the results need converting into units a human can measure — 47 teaspoons is arithmetically correct and practically useless. **Division with a remainder that must round up** appears with packaging: needing 3.2 bags of flour means buying four, and the leftover is real and has to be accounted for. **Division with a remainder that must round up again, but for a different reason**, appears with trays and batches: 7.4 trays means eight, and eight trays in an oven that holds three means three rounds of baking, which is where the time calculation begins.

Children meet remainders in exercises where the answer is a number. Here the answer is a decision, and the right decision depends on what is being divided — you round bags up, you round servings down, and you never round the oven. That distinction is the mathematical heart of the card and it is genuinely difficult.

The time calculation at the end usually produces the session's best moment, when a group realizes their plan requires the kitchen to start at four in the morning.

ACROSS K–5

Grades 4–5 run the arithmetic. Grades 2–3 handle the shopping list and the tally of bags. K–1 count people, count trays and are the ones who go and ask the office for the real number, which is a real errand with a real outcome.

EXTEND IT Actually cook for another class, at a scale the group has planned. Grades 4–5: add a budget constraint and find where the plan has to change.

WATCH OUT Keep this a planning exercise unless the school's kitchen staff are genuinely involved and willing — real food production for a school has regulatory requirements that are not the class's to shortcut.

THE THREAD UNDERNEATH The answer to a real problem is a decision, not a number.

The Potato in the Jug

Volume of a lumpy object, by displacement

How would you measure the volume of a potato?

Grades 3–5 50 min Real mess No heat needed Needs the kitchen

YOU WILL NEED

- Potatoes, apples, lemons, an egg, a carrot and a few other irregular items
- A large measuring jug with clear graduations, and a bowl to catch overflow
- A digital scale
- Centimeter cubes or squared paper, for the regular-solid comparison
- A recording table with columns for mass, volume and the two divided

WHAT YOU DO

1. Measure a rectangular block of something — a cheese block, a stick of butter — with a ruler and compute its volume.
2. Now try to do the same with a potato. Let the frustration land before offering anything.
3. Fill the jug to a marked line. Lower the potato in. Read the new line. Subtract.
4. Do this for every object and record mass and volume side by side.
5. Divide mass by volume for each. Sort the objects by the result.
6. Drop each object in a bowl of water. Compare which float to where they sit in your sorted list.

EVERYONE PARTICIPATES

Measuring, pouring, recording and calculating. Nothing is eaten, the produce is apparatus, and the water-and-jug work is calm and repetitive in a way that suits a lot of children.

WORTH ASKING

- Why can we not just use a ruler on the potato?
- What does the number you got from dividing actually tell you?
- Predict this one. Will it float? Why?

JOURNAL Mass, volume and the ratio, in a three-column table

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

An object pushes aside exactly its own volume of water, so the rise in the jug is the object's volume — no formula and no measuring of an awkward shape required. It is over two thousand years old as an idea and it still feels like a trick the first time a child sees it work.

Putting mass and volume side by side then gives density without ever using the word first. Divide one by the other and a number appears; sort the objects by that number and the ordering turns out to predict which ones float. Objects whose number comes out below about one — the value for water in grams per milliliter — float, and the ones above sink. Children who derive that from their own table own it in a way that being told the definition does not produce.

Two practical notes. Fresh potatoes usually sink and old ones sometimes do not, for the same reason the old egg floated in every density lesson ever taught — moisture has left. And anything with air trapped in it, like a floury apple, will misbehave, which is a finding rather than an error.

ACROSS K–5

K–1 do the lowering and the reading of the line, with an older partner steadying the jug. Grades 2–3 record. Grades 4–5 do the division, the sorting and the prediction of what will float before testing it.

EXTEND IT Compare a whole apple with the same apple cut into pieces — the volume should barely change and the children should say why. Grades 4–5: measure the volume of something that floats, using a weight to hold it under, and work out how to correct for the weight.

WATCH OUT Water and a hard floor make the same hazard every time. Trays under everything, a mop nearby, and an agreement that spills get announced rather than hidden. If a real egg is used, hands are washed immediately after handling it and the jug water changed — or use a hard-boiled egg, which behaves identically for this.

THE THREAD UNDERNEATH When the obvious method fails, the answer is usually to measure something else instead.

Watching Something Cool Down

Line graphs, rate of change, and a curve that flattens

Does hot tea cool at the same speed the whole way down?

Grades 3–5 55 min Light mess Adult at the heat Needs the kitchen

YOU WILL NEED

- Identical mugs or beakers, and hot water (adult poured)
- Thermometers, one per container
- A lid, a cloth cozy, a wide shallow dish and a narrow tall one — the variables
- Timers, and pre-ruled graph paper with time on the horizontal axis
- Colored pencils, one color per condition

WHAT YOU DO

1. Predict first, on paper: sketch the shape you think the cooling graph will be. Most children draw a straight line.
2. Adult pours equal volumes of hot water into four containers that differ in one way each.
3. Record every container's temperature every two minutes for thirty minutes. Nobody skips a reading.
4. Plot all four on one set of axes as you go, not at the end.
5. Compare the finished curves with the predictions everyone drew at the start.
6. Work out where each line was steepest, and say what that means in words.

EVERYONE PARTICIPATES

Hot water rather than a drink, deliberately — this is apparatus, not refreshment. Reading, recording, plotting and interpreting are the entire session.

WORTH ASKING

- You drew a straight line. What actually happened?
- Where is your line steepest? What was happening then?
- Which container would you use to keep something hot? Which to cool something fast?

JOURNAL Prediction sketch first, then the real four-line graph on the same page

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Something hot loses heat fastest when it is furthest from the temperature of the room, and more slowly as the gap narrows. So a cooling graph is steep at the start and flattens as it goes, approaching room temperature without ever quite arriving — and almost every child predicts a straight line, which makes this one of the best prediction-versus-data cards available.

The variables explain themselves once the curves are side by side. A lid stops evaporation, which carries away a great deal of heat; a wide shallow dish has far more surface in contact with the air than a tall narrow one and cools much faster; a cozy slows conduction outward. Same water, same starting temperature, four different curves.

Mathematically this is where a line graph earns its keep over a bar chart: the shape of the line is the information, and the steepness at a point is a rate. Asking children to say where the line is steepest and what that means in plain words — 'it was losing about three degrees a minute at the start and less than one degree a minute at the end' — is rate-of-change reasoning years before it is called that.

ACROSS K–5

K–2 read thermometers and call numbers on the two-minute beat, which is a genuinely useful job that keeps the whole experiment running. Grades 4–5 plot live, manage four lines at once and do the steepness analysis.

EXTEND IT Connect to the cold chain on card T03 — the same physics, running the other direction. Grades 4–5: predict how long until each container reaches room temperature and check against the extended curve.

WATCH OUT Adults pour all hot water, containers stay on trays at the back of the bench, and 'hot' here means 50°C or below, measured before pouring, in non-glass containers on trays. Nobody drinks anything.

THE THREAD UNDERNEATH The world mostly does not change at a steady rate, even though that is what we expect.

The Class Vote

Anonymous data, fair comparison, and what a number can honestly claim

Eleven people preferred it. Out of how many? And does that settle it?

Grades K-2 & 3-5 45 min Light mess No heat needed Needs the kitchen

YOU WILL NEED

- Two or three versions of something the class made, differing in one way
- A jar per option and a bowl of dried beans — one bean per person
- Or: dot stickers and a chart, which works equally well
- A screen or a box, so nobody can see where anyone drops their bean
- Chart paper and a calculator

WHAT YOU DO

1. Agree the question before voting. 'Which do you prefer' and 'which is sweeter' are different questions.
2. Everyone gets one bean. The jars are behind a screen so no vote is visible.
3. Anyone may vote on how it looked or smelled instead of how it tasted. Say this out loud; it is not a concession.
4. Count the beans in front of everyone. Build a bar chart.
5. Work out the fraction and the percentage. Say the denominator out loud every time you quote the result.
6. Then the hard question: is the gap big enough to mean anything? What would make you more confident?

EVERYONE PARTICIPATES

Everyone votes, on whatever basis they choose, and the ballot never records why. A vote cast on smell counts exactly as much as a vote cast on taste — and because it is anonymous, nobody in the room could tell the difference even if they wanted to.

WORTH ASKING

- Eleven out of how many?
- Would the result be different if we had asked forty people?
- Is nine to eight a difference, or is it nearly a coin toss?

JOURNAL Bar chart of the class result — the class's number, not yours

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

This card is about data handling and about honesty, and the two are the same thing here.

The **mechanics** are deliberate. Anonymous voting behind a screen means no child's choice is visible, which removes both peer pressure and the possibility of anyone being watched for whether they tasted. Bean-drop and dot-voting are standard tools in school food programs precisely because they are anonymous and produce a physical bar chart as they go. One vote each is the fairness rule and children enforce it fiercely.

The **mathematics** is proportional reasoning with a real denominator. 'Eleven liked it' means nothing; eleven out of fourteen and eleven out of forty are different claims, and children quote the first kind constantly. Insisting the denominator is said every single time is the single most useful data-handling habit in elementary school.

And then the honest ending: a nine-to-eight split is not a finding, it is close to a coin toss. Sitting with that rather than declaring a winner is the same lesson as the triangle test on card N04, and it inoculates children against a great deal of what they will be told by graphs later in life.

ACROSS K-5

K-1 drop beans and count them into tens, which is real place-value work. Grades 2-3 build the chart. Grades 4-5 compute percentages and lead the is-it-big-enough discussion.

EXTEND IT Ask another class the same question and compare. Grades 4-5: run the same vote twice on different days with the same people and see how much the answer moves on its own.

WATCH OUT Never announce or record who voted for what, and never say how many children tasted. The result is a class number. Collect the beans and start again cleanly, and do not carry results forward as a chart of anyone's progress.

THE THREAD UNDERNEATH A number can be true and still not prove what someone says it proves.

A Pinch, a Handful, a Gram

Non-standard and standard units, and why anyone bothered

Whose handful is the right handful?

Grades K-2 & 3-5 50 min Real mess No heat needed Classroom is fine

YOU WILL NEED

- Dried rice or lentils, in a big bowl
- A digital scale reading grams
- Teacups, tablespoons, and everyone's hands
- String, for measuring in handspans and forearms
- Graph paper for the class spread

WHAT YOU DO

1. Everyone takes one handful of rice and weighs it. Write every result on the board, no names.
2. Find the smallest, the largest and the middle. How big is the gap?
3. Do the same with a pinch, and with a teacup filled 'about full'.
4. Now everyone weighs out 50 g. Put those results on the board too.
5. Measure the room in handspans — each person their own — then in forearms, then with a meter stick.
6. Plot both spreads side by side and say, in one sentence, what standard units are for.

EVERYONE PARTICIPATES

Dried rice, scales and string. No food is prepared or eaten, and the tactile work with dry grains is calm and popular. Every child's handful goes on the board without a name attached.

WORTH ASKING

- Whose handful should the recipe mean?
- Why did everyone's 50 g agree?
- When would a handful be a perfectly good measure?

JOURNAL Two spreads on one page — handfuls, and weighed grams

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Non-standard units are not primitive; they are local. A handful, a pinch, a teacup and a forearm are all fine measures if the person measuring is the person cooking, and most of the world's cooking has always worked that way. The problem appears the moment a recipe has to travel to somebody else's hands.

The spread on the board makes that concrete. A class of mixed ages will produce handfuls ranging over a factor of two or more, and the weighed 50 g results will all agree. The difference between the two spreads is the argument for standardization — not a rule to be accepted but a conclusion the class reached from its own numbers.

It is also worth telling children how far this has been taken. The kilogram was defined for over a century by a single metal cylinder kept in a vault near Paris, and every other kilogram in the world was a copy of a copy of that object — which meant that if it changed, the kilogram changed. It was redefined in 2019 in terms of a constant of nature instead, so that anyone anywhere with the right equipment can reproduce it without borrowing anything from anyone. That is the same problem as the handful, solved at the largest possible scale.

ACROSS K-5

This card is at its best in a mixed-age room, because the spread of handfuls is real and visible and the youngest children's measurements matter to the argument. Say so explicitly — a kindergartener's handful is not a wrong handful, it is the data point that makes the point.

EXTEND IT Find recipes that still use non-standard measures and discuss who they were written for. Grades 4–5: compute the range and the mean of the class handfuls and express the spread as a percentage of the mean.

WATCH OUT Dried rice on a hard floor is a genuine slip hazard. Trays under everything, and sweep before anyone walks.

THE THREAD UNDERNEATH Agreeing on what a word means is what lets strangers work together.

The Dish That Traveled

A recipe with a person attached to it

Who made this before you were born?

Grades K-2 & 3-5 Two sessions plus home time Real mess Adult at the heat Needs the kitchen

YOU WILL NEED

- The interview recordings and notes from card T02
- Ingredients for two or three dishes families have offered, chosen with them
- A large map, and pins and string
- Recipe cards, to be written by hand
- An invitation for family members to come and cook alongside, if they want to

WHAT YOU DO

1. Each child names one dish that means something to someone they know — a family member, a neighbor, a member of school staff. Nobody brings food or ingredients; the class buys everything it cooks.
2. Pin the route the dish itself traveled — where its ingredients grow, where it is cooked now. Map the dish, never the family.
3. Choose two or three to cook together, in consultation with the family who offered them.
4. Invite that family member to cook with the class if they would like to. If they cannot come, play the recording.
5. Cook it their way, not a simplified version. Ask about every step you would have guessed at.
6. Write the recipe by hand, credited by name if the family wants that, and give a copy back to them.

EVERYONE PARTICIPATES

Cooking somebody's dish and eating it are separate things and the card only requires the first. A child cooks, serves, asks questions and writes the card, and whether they taste it is unremarked. Components are served separately as always, which also happens to be how a great many of these dishes are traditionally served.

WORTH ASKING

- Who taught the person who taught you?
- What would be lost if we changed this ingredient?
- What does this dish get made for?

JOURNAL The dish, drawn, with the map route and one sentence from the person

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Almost every dish has traveled. Ingredients moved by trade, by migration, by colonisation and by chance long before anyone wrote recipes down, and a map with string on it shows children something true that a textbook paragraph does not: that the food in their kitchen has a route.

The design rule that makes this work is that the **expert is the family member, not the teacher**. A recipe collected from a person, cooked their way, with their corrections, and given back to them credited, is a different act from a lesson about other people's food. It positions a parent who may not feel confident in a school building as the most knowledgeable person in the room, which is worth arranging on its own merits.

Two things to hold. Do not simplify or substitute without asking — 'we couldn't get that so we used something else' is a small thing to a school and can be a large thing to a family. And do not assign a child a culture; let every child choose whose dish they bring, which might be a grandparent's, a neighbor's, or the school cook's, and all three are equally valid.

ACROSS K-5

Pair a grade 4-5 scribe with a K-1 questioner. Older children handle heat and write the recipe; younger children ask the questions, which they do without embarrassment and which produces better answers.

EXTEND IT Build the class map across the whole term and display it at family night. Grades 4-5: research how one ingredient in the dish reached the place the dish comes from.

WATCH OUT Never require a family to participate or a child to produce a heritage. Some families will not want to, some have nothing they wish to share, and a child with no dish to bring should have an equally interesting alternative ready — the school cook, a local restaurant, a neighbor — arranged quietly in advance. Never map where a child's family is from: some families will not want that visible and a child cannot opt out of a wall. And any dish not in the term pack goes to every family, with its full ingredient list, at least a week before it is cooked.

THE THREAD UNDERNEATH Somebody's grandmother is the expert today, and the room should feel like that.

One Grain, Many Kitchens

The same staple, cooked completely differently

Half the world eats rice every day. Do they all eat the same thing?

Grades 3–5 70 min Real mess Adult at the heat Needs the kitchen

YOU WILL NEED

- Rice in several forms: long grain, short grain, basmati, arborio, glutinous, parboiled, flaked
- Or maize: whole kernels, polenta, masa, hominy — and wheat: berries, flour, semolina, bulgur
- Pans, a scale, water, a timer
- A map, and photographs of dishes from several places
- Small bowls for the side-by-side comparison

WHAT YOU DO

1. Look at and handle every form of the grain dry. Sort by size, shape, color, hardness.
2. Predict which will absorb the most water and which will take longest.
3. Cook three or four of them, weighing before and after and timing each.
4. Compare the cooked results side by side — sticky, separate, creamy, firm.
5. Match each to dishes and places on the map where that behavior is exactly what is wanted.
6. Work out why. Which property of the grain makes it right for that dish?

EVERYONE PARTICIPATES

Handling dry grains, weighing, timing and comparing textures make up the session. Cooked samples are compared by look and feel — stickiness, separateness, how a grain behaves under a fork — which is where the finding actually is. Taste if you want; nothing in the comparison needs it, and the texture vocabulary from card N01 does most of the work.

WORTH ASKING

- Which one absorbed the most? Was that what you predicted?
- Why would a cook want sticky rice? When would that be wrong?
- What grows where you live, and what is cooked with it?

JOURNAL Grain comparison page — dry drawing, cooked drawing, weights, words

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Rice is not one ingredient. Varieties differ in the proportions of two kinds of starch molecule, and that proportion largely determines whether cooked grains stay separate or cling together. Long-grain varieties cook up separate and dry; short-grain and glutinous varieties are much stickier; the varieties used for risotto release starch steadily under stirring, which is what makes the dish creamy without cream. None is better; each is right for what it is used for, and cuisines developed around the grain that grew locally.

Weighing before and after makes the absorption visible — grains roughly double or triple in weight, and the amount of water taken up varies by type, which is why a fixed water ratio fails when the variety changes.

The wider point is the one worth landing. A single staple crop supports enormous culinary diversity, and the diversity is not decorative — it reflects what grows in a place, what fuel and equipment were available, and what people wanted the food to do. Presenting it that way keeps the card from becoming a tour of other people's kitchens and makes it a lesson about materials and constraints.

ACROSS K-5

K-1 sort and handle dry grains, which is excellent sensory work with something completely unthreatening. Grades 2-3 weigh and time. Grades 4-5 run the absorption arithmetic and lead the matching of grain properties to dishes.

EXTEND IT Run the same card with maize or wheat and compare how many distinct foods come from one plant. Grades 4-5: express water absorption as a percentage of dry weight and rank the varieties.

WATCH OUT Boiling pans are adult-supervised throughout. Check for celiac disease and wheat allergy before running the wheat version, and note that some children avoid particular grains for religious reasons — ask in advance through the intake form rather than in the room.

THE THREAD UNDERNEATH Different is usually an answer to a different question, not a worse answer to the same one.

The Part You Were Going to Throw Away

Whole-ingredient cooking, stock, and what waste actually is

You used the broccoli. What is that in the trash?

Grades K-2 & 3-5 65 min Real mess Adult at the heat Needs the kitchen

YOU WILL NEED

- A normal session's worth of vegetable trimmings, saved deliberately
- Broccoli stalks, carrot tops, chard stems, herb stems, onion skins, parmesan rinds
- A large pot, a sieve, a scale, and containers to freeze stock in
- Two containers on a scale, labeled 'could be food' and 'could not'
- A peeler, a grater and a sharp knife

WHAT YOU DO

1. Weigh everything a normal prep session throws away. Write the number up.
2. Sort the trash into two piles: things that could be food and things that genuinely could not.
3. Weigh both piles. The first is usually far bigger than anyone guessed.
4. Cook from the first pile: broccoli-stalk slaw, carrot-top pesto, chard-stem pickle, herb-stem stock.
5. Simmer a stock from the rest, strain it, and freeze it in labeled portions for a later session.
6. Weigh what finally goes to compost. Compare with the first number and work out the percentage saved.

EVERYONE PARTICIPATES

Sorting, weighing, peeling and calculating. Nothing needs to be eaten, and the stock goes into the freezer for a later session rather than being served on the spot, which removes any expectation.

WORTH ASKING

- Why does this get thrown away when it is perfectly good?
- What percentage did we save?
- What is actually left in the pot after the stock is strained?

JOURNAL Before-and-after weights, and a drawing of the whole vegetable with every part labeled

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Most of what a kitchen throws away is edible, and the reasons it gets thrown away are habit, unfamiliarity and time rather than anything about the food. Broccoli stalks are sweeter than the florets once peeled; carrot tops are an herb; chard and beet stems are excellent pickled; onion skins give color and body to stock and dye to cloth. Demonstrating this with the class's own trash, weighed, is far more persuasive than a poster.

Stock is the technique that makes it systematic. Long gentle heating extracts soluble compounds — flavor, color, gelatine from bones, and savory body from a cheese rind — into water, and the solids are spent afterwards. It is an extraction, which is the same operation as making tea or the dye baths in card A01, and naming that connection is worth doing.

Keep the framing on **resourcefulness and skill** rather than on virtue or guilt. This is a card about knowing what things are and being able to cook, not about being good. It also has real money in it — the percentage saved is a number children can take home — and that is a more durable motivator than shame.

ACROSS K-5

K-2 sort the trash, which they are good at and enthusiastic about, and weigh. Grades 4-5 do the knife work on stalks and stems, manage the stock pot with an adult, and compute the percentage.

EXTEND IT Run the audit again at the end of term and compare. Grades 4-5: cost the saving over a year for the school kitchen — a real institution, with real numbers, and nobody's home in it.

WATCH OUT Only trimmings from that session's clean prep, never anything from plates or trash that has been handled or sat out. Stock pots are adult-managed and cooled quickly before freezing. Keep the tone free of guilt — some families in any class are managing food very carefully already and the last thing they need is a lecture.

THE THREAD UNDERNEATH Knowing what something is turns trash into ingredients. That is a skill, not a virtue.

Everyone Who Touched It

The people between a field and a plate

How many pairs of hands did this apple pass through?

Grades 3–5 60 min, plus a visit if you can arrange one Tidy No heat needed Classroom is fine

YOU WILL NEED

- One ordinary item with a long chain — an apple, a loaf, a carton of milk
- A long strip of paper across the wall, for the chain
- Cards and pins
- A visitor, if possible: a grower, a delivery driver, a shopkeeper, or the school's own kitchen staff
- Prepared questions from the interview work on card T02

WHAT YOU DO

1. Start with the apple on the table and work backwards, one step at a time, as far as the class can get.
2. Pin a card for every person, not every place — the point is people, and children default to places.
3. Count them. Then count again including the ones nobody mentioned: the person who made the crate, the one who fixed the truck.
4. Invite the school's own kitchen staff in first, because they are the closest link and are almost never asked.
5. Interview whoever visits using the technique from card T02. Add their real answers to the wall.
6. Ask each visitor the same closing question: what do you wish people knew about your work?

EVERYONE PARTICIPATES

No food and no tasting. This is interviewing, mapping and writing, and it is one of the cards that most reliably shifts a class's behavior outside the lesson.

WORTH ASKING

- How many people, really?
- Which ones did we forget the first time? Why those?
- What did they wish people knew?

JOURNAL The chain, drawn as a line of people, with one thing you learned from a visitor

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

A supply chain is a real object that children can reconstruct, and reconstructing it backwards is what makes it stick — from the plate to the shop to the truck to the packhouse to the picker to the grower to the person who grafted the tree. The exercise reliably produces a wall with twenty or more people on it for a single apple, and the surprise is the lesson.

Insisting on **people rather than places** is the design decision that matters. Left to themselves children produce a map; the card is about labor. Picking, sorting, driving, stacking and cooking are skilled and often hard jobs done by people who are largely invisible in the way food is usually discussed with children.

Starting with the school's own kitchen staff is deliberate and worth doing properly. They are in the building, they cook for these children daily, they are very rarely invited into a lesson as experts, and the relationship that follows changes how the dining hall feels for the rest of the year. Ask them first, give them notice, and let them decide what they want to talk about.

ACROSS K–5

K–2 draw the people for the wall and ask the direct questions nobody else dares to. Grades 4–5 research the invisible links, prepare the interview and write the thank-you letters — which should be real letters, sent.

EXTEND IT Cook something for the people on the wall who are in the building, and serve it to them. Grades 4–5: trace one imported ingredient and find out what is known — and what is not — about who grew it.

WATCH OUT Visitors need the school's usual safeguarding process and real notice. Do not set up a conversation in which children interrogate an adult about pay or conditions — let visitors say what they choose. Avoid any framing that ranks jobs.

THE THREAD UNDERNEATH Almost everything you eat was handled by someone you will never meet.

Cook It for Someone Else

A real recipient, a real delivery

Who in this building would be glad of something you made?

Grades K-2 & 3-5 Two sessions Real mess Adult at the heat Needs the kitchen

YOU WILL NEED

- A recipe the class can make reliably and scale — they should have made it before
- Packaging that travels, and labels listing every ingredient in full
- Card, for a handwritten note with each delivery
- A confirmed recipient, arranged in advance
- A cart or baskets

WHAT YOU DO

1. Decide together who it is for. Make the class choose — the choosing is half the session.
2. Find out what would actually suit them. Ask, rather than assuming.
3. Scale the recipe for the real number of people using card M02.
4. Cook, package, and label every single item with a complete ingredient list. No exceptions.
5. Write a note for each package, by hand, signed.
6. Deliver it in person if at all possible, and stay long enough to be thanked.

EVERYONE PARTICIPATES

Cooking for someone else is participation in food that requires no eating at all, and that is the point. A child who makes, packages, labels, writes the note and hands it over has done the entire thing. Serving others is also, not incidentally, one of the steps on the exposure ladder in card N02 — it happens here as an ordinary act of care rather than as an exercise.

WORTH ASKING

- How do we find out what they would actually like?
- What has to be on the label, and why is that not optional?
- What was it like to be thanked?

JOURNAL Who it was for, what you made, and what they said

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

This card exists because a class that only ever cooks for itself is doing an exercise, and the same class cooking for a named person who is expecting it is doing something with stakes. The change in care when the recipient is real and known is immediate and obvious — plating improves, labeling improves, and the arguing stops.

The obligations that come with feeding someone are the curriculum. **Ask first:** what would suit them, what can they not eat, when do they want it. **Label completely,** because a person receiving food from strangers cannot see what is in it and an allergy is not a preference — every item gets a full ingredient list and this is the one rule with no flexibility. **Deliver on time,** because a promise about food is a promise. **Accept thanks properly,** which children find surprisingly hard.

Choose recipients carefully. The strongest version is reciprocal and local — the kitchen staff, the office staff, the caretaker, a class of younger children. If food goes to a family, it is routed entirely through staff and the recipient is never named to the children. Anything that positions the children as benefactors and another group as needy does damage, and some of the children cooking may be from families in exactly the situation being discussed.

ACROSS K-5

Grades 4-5 scale and cook. K-2 package, write notes and deliver, which is the part with the most feeling in it. Put a younger child at the front for the handover.

EXTEND IT Make it a standing arrangement — the same recipients every month, so a relationship builds. Grades 4-5: ask for feedback and change the next batch in response to it.

WATCH OUT Follow the school's rules on food leaving the kitchen; some schools and districts restrict this significantly and it is far better to know before the children are invested. Full allergen labeling on everything, no exceptions. Avoid any framing that makes recipients into objects of charity.

THE THREAD UNDERNEATH Making something for a person who is expecting it is a different job from making it for practice.

How the World Eats

Utensils, table shapes and the rules nobody wrote down

Why do you hold your food the way you do?

Grades K-2 & 3-5 55 min Real mess No heat needed Needs the kitchen

YOU WILL NEED

- Eating tools: forks, chopsticks, spoons, a flatbread, a banana leaf or large plate, a bowl
- Foods suited to each: rice, noodles, something wrapped, something scooped — plus a bowl of stew that stays in its bowl as a demonstration of what a spoon is for
- Photographs of tables and shared meals from many places, used respectfully
- Paper, for the rules-of-the-table exercise
- Water and towels, because hands will be used

WHAT YOU DO

1. Look at several ways of eating and match tools to foods. Which tool suits which food, and why?
2. Try picking up and moving something with a tool you do not normally use — dry beans, rice, rolled paper, or food if you want food. Badly, at first, and that is fine.
3. Notice the engineering: chopsticks are a lever pair; a flatbread is a scoop you eat afterwards, a spoon is a small bowl on a stick.
4. Each child writes down three rules of their own family's table that nobody ever taught them out loud.
5. Rules go into a hat on unsigned slips and get read out at random. Some are shared, some are not, and none is the correct one. You may write a rule from any table you have eaten at.
6. Agree the rules for this class's own table, and write them up.

EVERYONE PARTICIPATES

Trying a tool is separate from eating with it, and a child can practice chopsticks on dry beans all session quite happily. The rules-of-the-table half involves no food whatever and is where the strongest conversation usually happens.

WORTH ASKING

- Why does this tool suit this food?
- What is a rule at your table that nobody ever said out loud?
- Whose rules should this class use?

JOURNAL Three rules from your table, and the class's agreed rules

WHAT IS ACTUALLY HAPPENING • FOR THE ADULT

Eating tools are engineering solutions shaped by the food, the fuel and the materials to hand. Chopsticks suit food already cut small, which suits cooking fast over scarce fuel. A flatbread that scoops is a utensil made of the meal. A spoon suits liquid, a fork suits food you must hold still and cut. Hands — the most common eating tool on earth — suit food at the right temperature and demand washing, which is why elaborate hand-washing customs accompany them.

The unwritten-rules exercise is the one that does the real work. Every family has table rules nobody stated: when you may start, whether you may leave, whether talking is expected, whether finishing is expected, what happens if you do not want something. Writing three of them down and comparing lists shows children that their own normal is one option among many — which is the actual content of the card, delivered by the children rather than by the teacher.

Handle the cultural material carefully. Show real meals, name places and people specifically rather than by continent, avoid presenting any way of eating as exotic, and give the children in the room whose families eat in these ways the first word if they want it — and no obligation if they do not.

ACROSS K-5

K-1 practice tools on dry materials, which is superb fine-motor work. Grades 4-5 lead the comparison of rules and chair the negotiation of the class's own table rules — and should be held to actually listening to the younger children's suggestions.

EXTEND IT Invite families to demonstrate how a dish is eaten as well as cooked. Grades 4-5: design an eating tool for a food that does not have a good one yet, using card E01's method.

WATCH OUT Never ask a child to perform their culture or to speak for a group. Eating with hands is normal for a great many people and should be presented that way, with proper hand-washing, not as a novelty. Check the photographs you use for anything that frames people as curiosities.

THE THREAD UNDERNEATH The way your family does it is one good way among many, and knowing that is a kind of freedom.